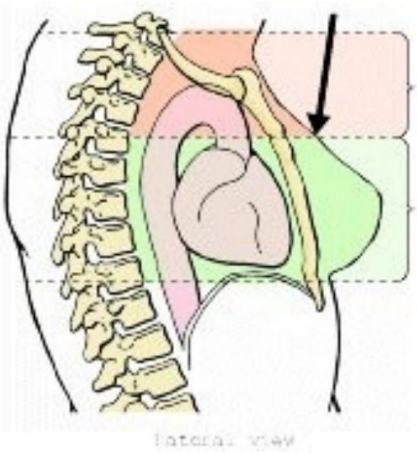
CP2 Anatomy: top 200 questions

Superior, anterior & posterior mediastinum

Mediastinum boundaries and subdivisions

1. In the following illustration, the arrows points to a line that represents the:



A. separation between the superior thoracic aperture and the mediastinum.

B. separation between the superior mediastinum and the inferior mediastinum.

C. separation between the superior and inferior thoracic lymph drainage areas.

D. separation between the axillary and abdominal lymph drainage of the breast.

E. level of the jugular notch

Answer: B. separation between the superior mediastinum and the inferior mediastinum.

Why: The line from the sternal angle to the T4–T5 disc is the **transverse thoracic plane**, separating **superior from inferior mediastinum**.

2. The sternal angle is an important landmark of the thorax. Which of the following events occurs at this level?

A. 3rd costal cartilages articulate with the sternum

B. Trachea bifurcates at this level

C. Apex of heart lies at this level

D. 6th thoracic vertebra is present at this level

E. SVC forms at this level

Answer: B. Trachea bifurcates at this level

Why: The **trachea bifurcates** at the sternal angle; the 2nd (not 3rd) cartilage attaches there and the level is T4–T5, not T6.

3. A 56-year-old man undergoes chest CT after a motor vehicle collision. Imaging reveals a mass located posterior to the pericardium, anterior to the thoracic vertebral bodies, and inferior to the transverse thoracic plane passing through the sternal angle and the T4–T5 intervertebral disc. The mass displaces the esophagus anteriorly but does not involve the pleural cavities. Based on the anatomical boundaries and subdivisions of the mediastinum, which of the following best describes the location of the mass?

A. Superior mediastinum, bounded inferiorly by the diaphragm and superiorly by the thoracic inlet

B. Anterior mediastinum, located between the sternum and the fibrous pericardium

C. Middle mediastinum, containing the heart and great vessels within the pericardial sac

D. Posterior mediastinum, located between the posterior surface of the pericardium and the thoracic vertebral bodies

E. Superior mediastinum, located between the manubrium and the bodies of the upper thoracic vertebrae

Answer: D. Posterior mediastinum, located between the posterior surface of the pericardium and the thoracic vertebral bodies

Why: Below the transverse thoracic plane and **behind the pericardium**, in front of the vertebrae, is the **posterior mediastinum**.

Superior mediastinum layout and thymus

4. From anterior to posterior, the correct order of structures in the superior mediastinum is:

A. Thoracic duct → Veins → Arteries → Trachea → Esophagus → Thymus

B. Thymus → Veins → Arteries → Trachea → Esophagus → Thoracic duct

C. Thymus → Trachea → Veins → Esophagus → Arteries → Thoracic duct

D. Arteries → Thymus → Thoracic duct → Esophagus → Trachea → Veins

Answer: B. Thymus → Veins → Arteries → Trachea → Esophagus → Thoracic duct

Why: Superior mediastinum front to back: **thymus, veins, arteries, trachea, esophagus, thoracic duct**.

Brachiocephalic veins and SVC

5. A thoracic CT scan demonstrates a mass immediately medial to the right lung compressing a vascular structure that lies anterolateral to the trachea and posterolateral to the ascending aorta. Which structure is most likely being compressed?

A. Aortic arch
B. Pulmonary trunk
C. Superior vena cava
D. Esophagus
E. Thoracic duct

Answer: C. Superior vena cava

Why: The **SVC** lies on the right of the superior mediastinum, **anterolateral to the trachea** and **posterolateral to the ascending aorta**.

6. Which of the following drains into the brachiocephalic veins?

A. Hemiazygos vein
B. Superior thyroid vein
C. Right superior intercostal vein
D. Left superior intercostal vein

Answer: D. Left superior intercostal vein

Why: The **left superior intercostal vein** crosses the arch to drain into the **left brachiocephalic vein**; the right one drains to the azygos.

7. A medical resident is preparing to insert a central venous line into the left brachiocephalic vein of her 10-year-old patient. Which of the following is a correct relation of the left brachiocephalic vein?

A. The vein lies posterior to the arch of the aorta
B. The brachiocephalic artery lies superficial to the vein
C. The vein crosses the anterior aspect of the manubrium of the sternum
D. The vein passes deep to the thymus gland
E. The apex of the left lung lies superficial to the vein

Answer: D. The vein passes deep to the thymus gland

Why: The left brachiocephalic vein passes behind the manubrium, **deep to the thymus** and anterior to the arch branches.

8. When performing a sternotomy, a resident goes too deep with his cut. Dark blood wells up before the sternal retractor can be applied. Which structure has most likely been injured during this procedure?

A. Left brachiocephalic v.
B. Arch of the azygos v.
C. Ascending aorta
D. Left subclavian v.
E. Left atrium

Answer: A. Left brachiocephalic v.

Why: Dark (venous) blood right behind the manubrium means the **left brachiocephalic vein**, which crosses just deep to the sternum.

9. A 65-year-old man presents with a swollen neck, marked edema in both upper limbs, and engorged and prominent intercostal veins. Subsequent examination reveals a tumor in his right lung. The tumor is likely compressing structures in which of the following regions?

A. Anterior mediastinum
B. Posterior mediastinum
C. Superior mediastinum
D. Hilum of the lung
E. Vena caval foramen of the diaphragm

Answer: C. Superior mediastinum

Why: Swollen neck, both arms and distended chest-wall veins = **SVC obstruction**, a structure of the **superior mediastinum**.

10. A 62-year-old man with a 45-pack-year smoking history comes to the clinic because his face and both arms have been swelling for 3 weeks. The swelling is worse when he bends forward. On exam, his face is flushed, the veins in his neck are distended, and dilated veins run across the front of his chest wall. A CT scan shows a mass in the right upper lobe extending into the right side of the superior mediastinum. Compression of which of the following structures best explains these findings?

A. Left brachiocephalic vein
B. Arch of azygos vein
C. Superior vena cava
D. Right subclavian vein
E. Right internal jugular vein

Answer: C. Superior vena cava

Why: Face and **both** arms swollen with distended neck/chest veins means the block is in the **SVC**, below where both brachiocephalics join.

Arch of the aorta and branches

11. During dissection of the superior mediastinum, a student identifies three major branches arising from the aortic arch. Which option correctly lists these branches from the patient's right to left?

- **A.** Left subclavian → left common carotid → brachiocephalic trunk
- **B.** Brachiocephalic trunk → left common carotid → left subclavian
- **C.** Brachiocephalic trunk → left subclavian → left common carotid
- **D.** Right common carotid → left common carotid → left subclavian

Answer: B. Brachiocephalic trunk → left common carotid → left subclavian

Why: Right to left the arch gives the **brachiocephalic trunk, left common carotid, left subclavian**.

12. A 34-year-old male is brought to the trauma center after sustaining a penetrating stab wound to the superior mediastinum just posterior to the manubrium sterni. Angiography reveals a transection of the first branch arising directly from the arch of the aorta. Based on the anatomical distribution of this specific vessel, perfusion will be compromised to which of the following combined anatomical regions?

- **A.** Left side of the head and neck only
- **B.** Left upper limb only
- **C.** Right side of the head and neck, and the right upper limb
- **D.** Right upper limb and left upper limb
- **E.** Left side of the head and neck, and the left upper limb

Answer: C. Right side of the head and neck, and the right upper limb

Why: The first arch branch is the **brachiocephalic trunk**; it splits into the right common carotid and right subclavian, so it feeds the **right head/neck and right upper limb**.

13. A 58-year-old aquarium diver was pierced in the chest by the barb of a stingray while transferring the animal to another tank. At the Emergency Department, examination found no sign of a pulse and a single puncture at the level of the manubriosternal joint, immediately adjacent to the sternum. Which structure has most likely been injured?

- **A.** Left brachiocephalic v.
- **B.** Aortic arch
- **C.** Inferior vena cava
- **D.** Right recurrent laryngeal n.
- **E.** Thoracic duct

Answer: B. Aortic arch

Why: A deep wound beside the sternum at the manubriosternal joint hits the **aortic arch**, which begins and ends at that level — pulseless exsanguination.

14. A 50-year-old heavy smoker was diagnosed with squamous cell carcinoma of the lung. His CT scan reveals a tumor located just posterior to the root of the left lung. Later, it was also determined that the mass is also at the level of T4 vertebrae. Blood flow in which of the following vessels is most likely compressed by this tumor?

- **A.** Supreme intercostal vein
- **B.** Arch of the aorta
- **C.** Left subclavian vein
- **D.** Ascending aorta
- **E.** Azygos vein

Answer: B. Arch of the aorta

Why: The arch descends **posterior to the root of the left lung** beside the **T4** vertebra, so a mass there compresses it.

Vagus and recurrent laryngeal nerves

15. In preparing for his Board Certification in thoracic surgery, a surgeon reviews the geographic relations of critical structures in the chest. Which of the following is correct regarding the relationships of the vagus nerves in their passage through the thorax?

- **A.** Right vagus passes onto the anterior aspect of the esophagus to become the anterior vagus
- **B.** Left vagus passes through the anterior mediastinum
- **C.** Right vagus passes posterior to the root of the lung
- **D.** Left vagus passes across the posterior side of the aortic arch
- **E.** Right vagus passes through the middle mediastinum

Answer: C. Right vagus passes posterior to the root of the lung

Why: Both vagi pass **posterior to the lung roots**; the right vagus becomes the posterior vagal trunk and crosses no anterior structures.

16. A 65-year-old man is complaining of hoarseness of voice. Patient reports being a chronic smoker for the past 20 years. He undergoes imaging for a suspected right apex lung tumor. Which nerve is most likely affected?

- **A.** Right phrenic nerve
- **B.** Right vagus nerve
- **C.** Right recurrent laryngeal nerve
- **D.** Left phrenic nerve
- **E.** Thoracic sympathetic trunk

Answer: C. Right recurrent laryngeal nerve

Why: The **right recurrent laryngeal nerve** hooks around the **right subclavian artery** at the lung apex, so a **right apical tumor** can compress it and cause **hoarseness**. A sympathetic-trunk hit would give Horner syndrome instead.

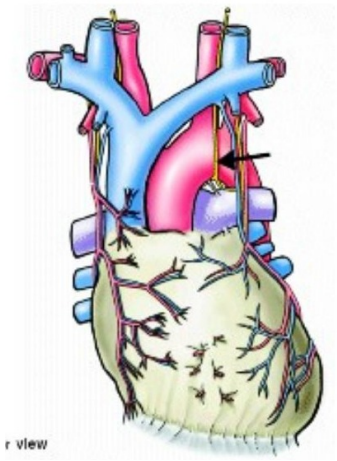
17. As the right vagus nerve enters the thorax anterior to the root of the neck, it gives off the right recurrent laryngeal nerve. Around which vessel does this nerve hook before ascending into the tracheoesophageal groove?

A. Arch of the aorta
B. Right subclavian artery
C. Brachiocephalic trunk
D. Right pulmonary artery
E. Internal thoracic artery

Answer: B. Right subclavian artery

Why: The **right RLN** hooks around the **right subclavian artery**; the **left RLN** hooks under the **aortic arch** by the ligamentum arteriosum.

18. In the following illustration, the arrow points to the:



A. left vagus nerve
B. left sympathetic nerve
C. left phrenic nerve
D. left internal thoracic artery
E. hemiazygos vein

Answer: A. left vagus nerve

Why: The nerve crossing the **left side of the aortic arch** toward the lung root is the **left vagus**.

19. A 62-year-old male patient expresses concern that his voice has changed over the preceding months. Imaging reveals a growth located within the aortic arch, adjacent to the left pulmonary artery. Which neural structure is most likely being compressed to cause the changes in the patient's voice?

A. Left phrenic nerve
B. Right phrenic nerve
C. Left recurrent laryngeal nerve
D. Right recurrent laryngeal nerve
E. Left vagus nerve
F. Right vagus nerve

Answer: C. Left recurrent laryngeal nerve

Why: A mass at the arch beside the left pulmonary artery compresses the **left RLN** where it hooks under the arch, causing hoarseness.

20. A 67-year-old man with a history of aortic arch aneurysm presents with progressive hoarseness. Laryngoscopy reveals paralysis of his left vocal fold. CT imaging demonstrates enlargement of the aortic arch with compression of a nerve that branches from the left vagus nerve, loops beneath the aortic arch near the ligamentum arteriosum, and then ascends toward the larynx. Which of the following best explains the anatomical basis of this patient's symptoms?

A. Compression of the left phrenic nerve as it passes anterior to the root of the left lung, resulting in paralysis of the left hemidiaphragm and impaired vocal fold movement.
B. Compression of the right recurrent laryngeal nerve as it loops beneath the right subclavian artery, resulting in paralysis of the left vocal fold.
C. Compression of the left vagus nerve as it passes anterior to the root of the left lung, directly interrupting the motor supply to the left vocal fold without affecting its recurrent branch.
D. Compression of the left recurrent laryngeal nerve as it loops beneath the aortic arch, disrupting motor innervation to most intrinsic muscles of the larynx and causing hoarseness.
E. Compression of the sympathetic trunk adjacent to the thoracic vertebral bodies, disrupting parasympathetic innervation to the larynx and causing vocal fold paralysis.

Answer: D. Compression of the left recurrent laryngeal nerve as it loops beneath the aortic arch, disrupting motor innervation to most intrinsic muscles of the larynx and causing hoarseness.

Why: The vocal-fold palsy comes from the **left RLN** looping under the arch, which supplies most **intrinsic laryngeal** muscles.

21. A patient has a patent ductus arteriosus and is scheduled for surgery to ligate the duct. During surgery, which of the following has the highest risk of being injured?

A. Left sympathetic chain
B. Left phrenic nerve
C. Left cardiac nerve
D. Left splanchnic nerve
E. Left recurrent laryngeal nerve

Answer: E. Left recurrent laryngeal nerve

Why: The ductus/ligamentum arteriosum is hooked by the **left recurrent laryngeal nerve**, so PDA ligation endangers it.

22. A 1-year-old child, born full term, has been having difficulty breathing, a rapid heart rate, and is not gaining weight as expected. The child is diagnosed with a large patent ductus arteriosus. NSAIDs did not work to close the patency and the decision is made to do a surgical closure. Which structure, closely associated with the ductus arteriosus, should be identified to prevent it from being damaged?

A. Ligamentum venosum
B. Right vagus n.
C. Left phrenic n.
D. Left recurrent laryngeal n.
E. Right recurrent laryngeal n.

Answer: D. Left recurrent laryngeal n.

Why: The **left RLN** passes under the arch **posterior to the ductus**/ligamentum arteriosum and must be found before closing it.

Phrenic nerves and sympathetic trunk

23. A 42-year-old woman undergoes surgery to remove a tumor from the superior mediastinum. During dissection, the surgeon identifies the thymic remnant anteriorly, the great vessels nearby, and the trachea and esophagus posteriorly. The surgeon must avoid injuring a nerve that passes anterior to the root of the lung and another nerve that passes posterior to the root of the lung. The surgeon also identifies the thoracic duct and azygos vein in the posterior mediastinal region. Which of the following correctly describes the anatomical relationships of these structures?

A. The phrenic nerve passes posterior to the root of the lung, while the vagus nerve passes anterior to it; the thoracic duct ascends to the right of the azygos vein.

B. The phrenic nerve passes anterior to the root of the lung, while the vagus nerve passes posterior to it; the azygos vein ascends on the right side of the vertebral column, and the thoracic duct ascends through the posterior mediastinum.

C. The sympathetic chains lie anterior to the trachea, while the esophagus lies anterior to the heart; the vagus nerve travels with the phrenic nerve anterior to the lung root.

D. The thymus lies posterior to the esophagus, while the heart lies superior to the trachea; the thoracic duct ascends through the middle mediastinum anterior to the pericardium.

E. The phrenic and vagus nerves both pass anterior to the root of the lung; the azygos vein ascends on the left side of the vertebral column, and the thoracic duct lies anterior to the trachea.

Answer: B. The phrenic nerve passes anterior to the root of the lung, while the vagus nerve passes posterior to it; the azygos vein ascends on the right side of the vertebral column, and the thoracic duct ascends through the posterior mediastinum.

Why: Phrenic **anterior** and vagus **posterior** to the lung root; azygos on the **right** of the column; thoracic duct in the posterior mediastinum.

24. A 25-year-old woman is admitted to hospital with dyspnea. Radiologic examination reveals a tumor invading the lung surface just anterior to the hilum. Which nerve is being compressed by the tumor?

A. Phrenic

B. Vagus

C. Intercostal

D. Recurrent laryngeal

E. Cardiopulmonary

Answer: A. Phrenic

Why: The **phrenic** nerve passes **anterior to the hilum**; compressing it paralyses the hemidiaphragm, causing dyspnea.

25. A 54-year-old woman undergoes surgery to remove a mass adjacent to the fibrous pericardium. During the procedure, the surgeon identifies a nerve descending along the lateral surface of the pericardium, anterior to the root of the lung. Postoperatively, the patient experiences shortness of breath, and a chest radiograph reveals elevation of the ipsilateral hemidiaphragm. Which of the following best explains the nerve injury and its consequences?

A. The vagus nerve was injured; it passes anterior to the root of the lung and provides the primary motor innervation to the diaphragm.

B. The sympathetic trunk was injured; it descends along the vertebral bodies and provides motor innervation to the diaphragm.

C. The phrenic nerve was injured; it descends anterior to the root of the lung and provides motor innervation to the ipsilateral hemidiaphragm, as well as sensory innervation to the fibrous pericardium and central diaphragmatic pleura.

D. The recurrent laryngeal nerve was injured; it descends along the lateral pericardium and supplies the diaphragm and mediastinal pleura.

E. The intercostal nerve was injured; it travels through the middle mediastinum and provides the primary motor innervation to the diaphragm.

Answer: C. The phrenic nerve was injured; it descends anterior to the root of the lung and provides motor innervation to the ipsilateral hemidiaphragm, as well as sensory innervation to the fibrous pericardium and central diaphragmatic pleura.

Why: A nerve on the lateral pericardium **anterior to the lung root** is the **phrenic**; injury elevates that hemidiaphragm.

26. A student identifies the right phrenic nerve in the cadaver. Which correctly labels the route this nerve travels?

A. Arises from C1–C3, descends posterior to the right lung root, and traverses the esophageal hiatus (T10) alongside the vagus nerve.

B. Arises from C3–C5, descends posterior to the right lung root, lateral to the left atrium, and exits via the aortic hiatus (T12).

C. Arises from C3–C5, descends lateral to the superior vena cava and right atrium, anterior to the right root of the lung, and traverses the diaphragm via the caval opening (T8) with the inferior vena cava.

D. Arises from C5–T1, pierces directly through the muscle belly of the anterior scalene, passes anterior to the subclavian vein, medial to the aortic arch, posterior to the right lung root, and traverses the median arcuate ligament.

Answer: C. Arises from C3–C5, descends lateral to the superior vena cava and right atrium, anterior to the right root of the lung, and traverses the diaphragm via the caval opening (T8) with the inferior vena cava.

Why: The right phrenic (**C3–C5**) runs **lateral to the SVC and right atrium**, **anterior to the lung root**, and reaches the diaphragm through the **caval opening (T8)** with the IVC. Every other option puts it posterior to the root.

27. Referred pain from biliary colic caused by gallstones would most likely be felt in which location?

A. Right medial arm

B. Around the umbilicus

C. Over right shoulder

D. Midline between the scapulae

Answer: C. Over right shoulder

Why: Gallbladder irritation of the diaphragm travels in the **phrenic** nerve (C3–C5), so pain is referred to the **right shoulder**.

28. A patient presents to the ER with difficulty breathing and complaints of right shoulder pain with use of accessory muscles. He had undergone cardiothoracic surgery two days ago. What artery runs with the most likely damaged nerve?

A. Internal thoracic

B. Pericardiacophrenic

C. Recurrent laryngeal

D. Subclavian

Answer: B. Pericardiacophrenic

Why: Dyspnea plus right shoulder pain means **right phrenic** injury, and the phrenic runs with the **pericardiacophrenic** artery.

29. A 45-year-old woman reports to her physician with chronic cough, chest pain, generalized weakness, fatigue and loss of appetite since 4 weeks. She notices that she has quite a significant weight loss. Physical examination shows mild degree of drooping of her right upper eyelid and constriction of right pupil. MRI shows a tumor in the apical segment of the upper lobe of the right lung. Based on the signs and symptoms and the MRI study, which structure is likely to be compressed resulting in the signs?

A. Thoracic duct

B. Arch of aorta

C. Azygos vein

D. Sympathetic chain

E. Vagus nerve

Answer: D. Sympathetic chain

Why: Ptosis and miosis on the right with an apical tumor = Horner's from compression of the **sympathetic chain**.

Trachea, esophagus and superior lymphatics

30. A patient has dysphagia due to compression of the esophagus within the superior mediastinum. Which structure normally compresses the esophagus anteriorly at this location?

A. Ascending aorta

B. Trachea

C. Azygos vein

D. Thoracic duct

E. Left recurrent laryngeal nerve

Answer: B. Trachea

Why: In the superior mediastinum the **trachea** lies directly **anterior** to the esophagus, pressing on its front.

31. A 15-year-old female overdosed on ibuprofen. At the emergency department, the decision was made to insert a nasogastric tube and administer activated charcoal. As the nasogastric tube passes into the thoracic portion of the esophagus, where might one expect resistance from constriction of the organ?

A. Aortic arch, left primary bronchus, thoracic diaphragm

B. Arch of the azygos v., right primary bronchus, left atrium

C. Pulmonary trunk, thoracic duct, thymus

D. Arch of the azygos v., superior lobe of right lung, thoracic diaphragm

Answer: A. Aortic arch, left primary bronchus, thoracic diaphragm

Why: The thoracic esophagus is narrowed by the **aortic arch**, the **left main bronchus**, and the **diaphragm** (esophageal hiatus).

Inferior and anterior mediastinum

32. A 28-year-old female presents for evaluation of an asymptomatic mass discovered on routine chest radiography. CT imaging reveals a well-circumscribed soft-tissue mass located within the anterior mediastinum. A thoracic surgeon plans a minimally invasive biopsy and reviews the exact structural boundaries defining this specific mediastinal space. Which of the following options correctly identifies the anatomical boundaries of the anterior mediastinum?

A. Anteriorly by the sternum and transversus thoracis muscles; posteriorly by the pericardium; superiorly by the sternal angle; inferiorly by the diaphragm

B. Anteriorly by the manubrium sterni; posteriorly by the T1–T4 thoracic vertebral bodies; superiorly by the superior thoracic aperture; inferiorly by the sternal angle

C. Anteriorly by the pericardium; posteriorly by the bodies of the T5–T12 thoracic vertebrae; superiorly by the transverse thoracic plane; inferiorly by the diaphragm

D. Anteriorly by the body of the sternum; posteriorly by the esophagus and trachea; superiorly by the 1st costal cartilage; inferiorly by the xiphisternal joint

E. Anteriorly by the costal cartilages; posteriorly by the ascending aorta; superiorly by the jugular notch; inferiorly by the central tendon of the diaphragm

Answer: A. Anteriorly by the sternum and transversus thoracis muscles; posteriorly by the pericardium; superiorly by the sternal angle; inferiorly by the diaphragm

Why: The **anterior mediastinum** lies between the **sternum + transversus thoracis** in front and the **pericardium** behind, from the **sternal angle** down to the **diaphragm**. Option C describes the posterior mediastinum; B the superior.

Posterior mediastinum and thoracic aorta

33. Which of the following structures and its location are correct?

A. Descending aorta - posterior mediastinum

B. Trachea - Anterior mediastinum

C. Thymus - middle mediastinum

D. Left subclavian - posterior mediastinum

E. Arch of aorta - middle mediastinum

Answer: A. Descending aorta - posterior mediastinum

Why: The **descending aorta** is a posterior mediastinal structure (DATES); trachea, thymus, left subclavian and arch are superior.

34. A 68-year-old male presents to the emergency department with sudden-onset, severe, tearing chest pain that radiates to his interscapular back. Contrast-enhanced CT angiography reveals an acute aortic dissection originating just distal to the left subclavian artery origin and extending inferiorly toward the diaphragm within the thoracic descending aorta. Anatomically, within which mediastinal compartment is this involved section of the aorta located?

A. Anterior mediastinum

B. Superior mediastinum

C. Middle mediastinum

D. Posterior mediastinum

E. Superior thoracic aperture

Answer: D. Posterior mediastinum

Why: The question asks about the **descending thoracic aorta** the dissection runs along, and that lies in the **posterior mediastinum**. Ascending aorta = middle; arch = superior. (N15-41-1 asks about the point of origin instead, which is still the arch.)

35. During a posterior mediastinal dissection along the anterolateral surface of the thoracic vertebral bodies, a surgeon isolates presynaptic sympathetic fibers originating from the T5 through T9 sympathetic ganglia. These branches descend medially and unite to form a distinct nerve that pierces the crus of the diaphragm to terminate in the celiac plexus. Which of the following nerves is composed of these specific spinal cord levels?

A. Least splanchnic nerve

B. Lumbar splanchnic nerve

C. Greater splanchnic nerve

D. Lesser splanchnic nerve

E. Cardiopulmonary splanchnic nerve

Answer: C. Greater splanchnic nerve

Why: Splanchnic levels: **greater T5–T9**, **lesser T10–T11**, **least T12**. Greater and lesser run in the posterior mediastinum (the S of DATES).

Azygos system of veins

36. A 60-year-old man undergoes chest CT to evaluate collateral venous circulation after obstruction of the superior vena cava. Imaging reveals enlargement of the azygos vein and several communicating veins along the posterior thoracic wall. The radiologist explains that these vessels provide an alternate route for venous blood to reach the heart. Which of the following best describes the anatomy and clinical significance of the azygos venous system?

A. The azygos vein ascends on the left side of the vertebral column and drains directly into the inferior vena cava, providing collateral flow around inferior vena cava obstruction.

B. The hemiazygos vein ascends on the right side of the vertebral column and drains directly into the superior vena cava, while the azygos vein drains the left posterior intercostal spaces.

C. The azygos vein ascends on the right side of the vertebral column, receives posterior intercostal veins, and arches over the root of the right lung to drain into the superior vena cava.

D. The accessory hemiazygos vein drains the lower right posterior intercostal spaces and crosses to the left to join the azygos vein near the diaphragm.

E. The azygos vein runs anterior to the heart within the middle mediastinum and primarily drains the anterior intercostal veins into the pulmonary veins.

Answer: C. The azygos vein ascends on the right side of the vertebral column, receives posterior intercostal veins, and arches over the root of the right lung to drain into the superior vena cava.

Why: The azygos ascends on the **right**, collects posterior intercostals, and **arches over the right lung root** into the SVC.

37. A patient has a mediastinal mass compressing the vessel that arches over the root of the right lung before entering the superior vena cava. Which function would most likely be impaired?

A. Drainage of the myocardium

B. Drainage of the posterior thoracic and thoracoabdominal walls

C. Drainage of the brain

D. Drainage of the upper limb only

E. Drainage of the liver

Answer: B. Drainage of the posterior thoracic and thoracoabdominal walls

Why: The vessel arching over the right lung root is the **azygos**, which drains the **posterior thoracic and thoracoabdominal walls**.

38. A surgeon is operating on the posterior thoracic wall and encounters a vein draining blood from the left 9th–11th intercostal spaces. Which pathway does this blood most likely follow?

A. Left 9th–11th posterior intercostal veins → accessory hemiazygos vein → azygos vein → SVC

B. Left 9th–11th posterior intercostal veins → hemiazygos vein → azygos vein → SVC

C. Left 9th–11th posterior intercostal veins → azygos vein → SVC

D. Left 9th–11th posterior intercostal veins → internal thoracic vein → brachiocephalic vein → SVC

E. Left 9th–11th posterior intercostal veins → IVC

Answer: B. Left 9th–11th posterior intercostal veins → hemiazygos vein → azygos vein → SVC

Why: Left 9th–11th intercostals → **hemiazygos** → azygos → SVC; the accessory hemiazygos takes 4th–8th.

39. During a radiological examination of the thorax, a radiologist identifies the point at which the hemiazygous vein connects the the azygous vein. This junction is most commonly found at the level of which thoracic vertebra?

A. T3

B. T6

C. T9

D. T12

Answer: C. T9

Why: The **hemiazygos** crosses behind the aorta to join the azygos at about **T9**.

40. During dissection of the posterior mediastinum, a medical student identifies the azygos vein arching over the root of the right lung before entering the superior vena cava. Which venous structure typically crosses the vertebral column at the level of T8, posterior to the descending aorta, esophagus, and thoracic duct, to drain into the azygos vein?

A. Internal thoracic v.

B. Hemiazygos v.

C. Accessory hemiazygos v.

D. Left superior intercostal v.

E. Superior epigastric v.

Answer: C. Accessory hemiazygos v.

Why: The **accessory hemiazygos** crosses at **T7–T8**; the hemiazygos crosses lower, at T9.

41. A 46-year-old patient comes to his doctor's office and complains of chest pain and headache. His computed tomography (CT) scan reveals a tumor located just superior to the root of the right lung. Blood flow in which of the following veins is most likely blocked by this tumor?

A. Hemiazygos vein

B. Arch of the azygos vein

C. Right subclavian vein

D. Right brachiocephalic vein

E. Accessory hemiazygos vein

Answer: B. Arch of the azygos vein

Why: Just **above the right lung root** lies the **arch of the azygos**, hooking forward into the SVC.

42. A bronchoscopic procedure goes wrong and results in a puncture of the posterior wall of the right main bronchus filling the instrument with a slow ooze of blood. Which blood vessel is most likely damaged in this procedure?

A. Left brachiocephalic vein

B. Azygos vein

C. Brachiocephalic artery

D. Left subclavian artery

E. Thoracic aorta

Answer: B. Azygos vein

Why: The **azygos** lies behind and arches over the **right main bronchus**; a slow venous ooze means a vein.

Esophagus in posterior mediastinum

43. A 35-year-old man is admitted to the hospital with pain on swallowing. Imaging reveals a dilated left atrium. Which structure is most likely being compressed by the expansion of the left atrium to result in the patient's symptoms?

A. Esophagus

B. Root of the lung

C. Trachea

D. Superior vena cava

E. Inferior vena cava

Answer: A. Esophagus

Why: The esophagus lies just **behind the left atrium**, so a dilated atrium compresses it and causes painful swallowing.

44. During a surgical repair of a hiatal hernia, the surgeon must be careful not to damage the nerve fibers traveling alongside the esophagus through the diaphragm. Which of the following structures normally accompanies the esophagus through its diaphragmatic opening?

A. Azygos vein

B. Anterior and posterior vagal trunks

C. Right phrenic nerve

D. Thoracic duct

E. Greater splanchnic nerve

Answer: B. Anterior and posterior vagal trunks

Why: The **anterior and posterior vagal trunks** pass with the esophagus through the esophageal hiatus.

Thoracic duct and right lymphatic duct

45. During a procedure involving the left side of the thorax, a lymphatic vessel is traced toward the duct that normally drains lymph from the left thorax into the venous circulation. Where does this duct empty?

A. Junction of the right internal jugular and right subclavian veins

B. Right atrium

C. Junction of the left internal jugular and left subclavian veins

D. Pulmonary trunk

E. Azygos vein

Answer: C. Junction of the left internal jugular and left subclavian veins

Why: The thoracic duct empties into the **left venous angle**: left internal jugular + left subclavian junction.

46. A 54-year-old female undergoes surgical resection and regional lymph node dissection for a right-sided thoracic tumor. Postoperatively, she develops severe non-pitting lymphedema localized strictly to her right upper extremity, right side of the head and neck, and right chest wall. Physical examination reveals no edema in the left upper extremity, abdomen, or lower extremities. Obstruction of which of the following structures accounts for this localized presentation, and what anatomical organization explains why the left upper extremity and lower body are completely spared?

A. Thoracic duct obstruction; it receives lymph exclusively from the right upper quadrant superior to the diaphragm

B. Bronchomediastinal trunk transection; it drains the right arm and right chest wall while the left side drains via the cisterna chyli

C. Right lymphatic duct obstruction; it drains only the right upper quadrant of the body, whereas the thoracic duct drains the remaining three-quarters of the body

D. Cisterna chyli disruption; it receives lymph from the right upper limb directly before ascending into the superior mediastinum

E. Right internal jugular trunk occlusion; it drains the right upper extremity, chest wall, and lower extremities, leaving the left upper quadrant intact

Answer: C. Right lymphatic duct obstruction; it drains only the right upper quadrant of the body, whereas the thoracic duct drains the remaining three-quarters of the body

Why: Swelling only in the **right arm, right head/neck and right chest** = **right lymphatic duct** territory. The **thoracic duct** drains the other three-quarters, so those areas are spared.

47. During exploratory surgery, you see the surgeon elevate the esophagus and look between the descending aorta and azygos vein. What structure was she looking for?

A. greater thoracic splanchnic

B. left recurrent laryngeal

C. right pulmonary

D. sympathetic trunk

E. thoracic duct

Answer: E. thoracic duct

Why: The **thoracic duct** lies behind the esophagus **between the descending aorta and azygos** vein.

48. A 65-year-old, overweight male, with a history of smoking and GERD, has been experiencing worsening acid reflux over the past three weeks. Two years ago, he was diagnosed with Barrett's esophagus. Endoscopy reveals dysplasia (precancerous changes). He undergoes a surgical procedure to remove the distal portion of the esophagus. A post-operative radiograph shows a hazy opacity of the mediastinum, and a CT shows fluid collecting in the posterior mediastinum. Which structure was most likely damaged during surgical procedure?

A. Azygos v.

B. Descending aorta

C. Right bronchial a.

D. Thoracic duct

Answer: D. Thoracic duct

Why: The **thoracic duct** lies just behind the distal esophagus; fluid in the posterior mediastinum after esophagectomy is chyle.

Diaphragm passages and clinical correlates

49. A 62-year-old man goes to the Emergency Department with a history of severe dysphagia and retrosternal burning pain. During an endoscopy, the endoscope is advanced until the tip reaches the esophageal hiatus of the thoracic diaphragm. At which vertebral level did the tip of the scope most likely reach?

A. T8
B. T12
C. T7
D. T11
E. T10

Answer: E. T10

Why: The **esophageal hiatus** lies at **T10** (caval T8, aortic T12).

50. A 48-year-old man undergoes imaging for progressive difficulty swallowing. CT reveals an enlarged lymph node in the posterior mediastinum that compresses the esophagus near the level where it passes through the diaphragm. The patient is also found to have a dilated azygos vein and impaired lymphatic drainage from the lower body. A medical student reviews the major pathways through the diaphragm to determine which structures may be affected. Which of the following correctly matches a structure with its diaphragmatic passage?

A. The inferior vena cava passes through the esophageal hiatus at T10, accompanied by the vagal trunks.
B. The esophagus passes through the caval opening at T8, accompanied by the thoracic duct.
C. The thoracic duct passes through the aortic hiatus at T12, posterior to the diaphragm, while the esophagus passes through the muscular esophageal hiatus at T10.
D. The azygos vein passes through the caval opening at T8, accompanied by the right phrenic nerve.
E. The sympathetic trunks pass through the esophageal hiatus at T10, accompanied by the esophagus and vagal trunks.

Answer: C. The thoracic duct passes through the aortic hiatus at T12, posterior to the diaphragm, while the esophagus passes through the muscular esophageal hiatus at T10.

Why: The **thoracic duct** passes through the **aortic hiatus (T12)** behind the diaphragm; the esophagus uses the muscular hiatus at T10.

51. Which of the following structures pass posterior to the diaphragm?

A. Descending Aorta
B. Esophagus
C. Inferior Vena Cava
D. Right Phrenic nerve

Answer: A. Descending Aorta

Why: The **aorta** passes **behind** the diaphragm (aortic hiatus, T12) with the thoracic duct. The esophagus, IVC and phrenic nerves **pierce** it.

52. A 63-year-old man presents with progressive shortness of breath, facial swelling, distended veins in his neck, and persistent hoarseness. CT imaging reveals a large mass in the superior mediastinum that compresses the superior vena cava and the left recurrent laryngeal nerve. Further imaging shows enlarged lymph nodes extending into the posterior mediastinum, where they displace the esophagus. Which of the following best explains the anatomical basis of this patient's clinical findings?

A. Compression of the inferior vena cava causes facial swelling, while injury to the right phrenic nerve produces hoarseness and esophageal displacement.
B. Compression of the superior vena cava impairs venous drainage from the head and upper extremities, compression of the left recurrent laryngeal nerve causes hoarseness, and posterior mediastinal involvement can produce dysphagia through esophageal compression.
C. Compression of the thoracic duct causes distended neck veins, while compression of the vagus nerve anterior to the lung root directly paralyzes the left vocal fold.
D. Compression of the azygos vein alone causes upper-body venous congestion, while posterior mediastinal involvement produces hoarseness by compressing the sympathetic trunks.
E. Compression of the trachea causes facial and neck swelling, while compression of the thoracic sympathetic chain produces vocal fold paralysis and dysphagia.

Answer: B. Compression of the superior vena cava impairs venous drainage from the head and upper extremities, compression of the left recurrent laryngeal nerve causes hoarseness, and posterior mediastinal involvement can produce dysphagia through esophageal compression.

Why: SVC compression swells head and arms, **left RLN** compression causes hoarseness, and posterior nodes push on the **esophagus** (dysphagia).

Middle mediastinum & the heart

Middle mediastinum contents

53. Which of the following correctly describes the anatomical relationship of the pulmonary trunk to the ascending aorta?

A. It lies posterior and to the right of the ascending aorta

B. It lies anterior and to the left of the ascending aorta

C. It lies posterior and to the left of the ascending aorta

D. It lies anterior and to the right of the ascending aorta

Answer: B. It lies anterior and to the left of the ascending aorta

Why: The pulmonary trunk leaves the RV and runs **anterior and to the left** of the ascending aorta before splitting into right and left pulmonary arteries.

54. During dissection, a student identifies a vessel branching from the pulmonary trunk and traveling toward the right lung. The vessel passes posterior to the ascending aorta and superior vena cava. Which vessel is being identified?

A. Right pulmonary artery

B. Left pulmonary artery

C. Right superior pulmonary vein

D. Azygos vein

Answer: A. Right pulmonary artery

Why: The **right pulmonary artery** runs horizontally to the right lung, passing **posterior to the ascending aorta and SVC**.

Pericardial sinuses and pericardial supply

55. A 28-year-old man is brought to the emergency department after sustaining a penetrating injury to the chest. Echocardiography reveals fluid accumulating around the heart, raising concern for cardiac tamponade. Which of the following correctly describes the organization of the pericardium and the location of the pericardial cavity?

A. The fibrous pericardium lies internally, followed by the visceral serous pericardium and then the parietal serous pericardium; the pericardial cavity lies between the fibrous and parietal layers.

B. The visceral serous pericardium forms the outermost layer, the fibrous pericardium directly covers the myocardium, and the pericardial cavity lies between the epicardium and myocardium.

C. The parietal serous pericardium covers the myocardium, while the visceral serous pericardium lines the inner surface of the fibrous pericardium; the pericardial cavity lies inside the myocardium.

D. The fibrous pericardium forms the outer layer, while the parietal and visceral serous layers fuse permanently; the pericardial cavity lies between the myocardium and endocardium.

E. The fibrous pericardium forms the outer layer, the parietal serous pericardium lines its inner surface, and the visceral serous pericardium covers the heart; the pericardial cavity lies between the parietal and visceral serous layers.

Answer: E. The fibrous pericardium forms the outer layer, the parietal serous pericardium lines its inner surface, and the visceral serous pericardium covers the heart; the pericardial cavity lies between the parietal and visceral serous layers.

Why: Outer **fibrous** layer, lined by **parietal serous** layer; **visceral serous** covers the heart. The pericardial cavity lies between the two serous layers.

56. A 39-year-old man is admitted to the hospital with a complaint of severe retrosternal pain that radiates to the left shoulder. The pain is relieved by leaning forward. Auscultation reveals a pericardial friction rub, leading to a diagnosis of pericarditis. Which of the following nerves is responsible for the radiating pain to the shoulder?

A. Intercostobrachial

B. Phrenic

C. Long thoracic

D. Greater thoracic splanchnic

E. Cardiopulmonary

Answer: B. Phrenic

Why: The **phrenic nerve (C3–C5)** carries pericardial pain; C3–C5 also supply the shoulder skin, so the pain is referred there.

57. Which of the following is incorrect pertaining to the pericardium?

A. It consists of visceral and parietal layers of serous pericardium, and the fibrous pericardium.

B. The visceral and parietal layers of the serous pericardium are continuous around the aorta and pulmonary trunk where they exit the heart.

C. It is mainly supplied with blood from the pericardiophrenic artery.

D. It has pain fibers that are conveyed by the intercostal nerves.

E. It encloses the terminal part of the inferior vena cava.

Answer: D. It has pain fibers that are conveyed by the intercostal nerves.

Why: Pericardial pain travels in the **phrenic nerves**, not intercostal nerves; supply is from pericardiacophrenic arteries and the sac encloses the terminal IVC.

58. During cardiopulmonary bypass, a surgeon passes a clamp behind the ascending aorta and pulmonary trunk to temporarily control blood leaving the heart. Through which space is the clamp passed?

A. Oblique pericardial sinus

B. Transverse pericardial sinus

C. Coronary sinus

D. Costomediastinal recess

E. Superior mediastinum

Answer: B. Transverse pericardial sinus

Why: The **transverse pericardial sinus** lies behind the ascending aorta and pulmonary trunk, letting a clamp pass around both.

59. During an open chest surgery, the surgeon gently lifted the apex of the heart and identified a blind sac posterior to the heart, which of the following vessels form the border of this blind sac?

A. Pulmonary vein + Inferior vena cava + pericardium

B. Pulmonary vein + Pulmonary artery + pericardium

C. Pulmonary artery + Inferior vena cava + pericardium

D. Inferior vena cava + Superior vena cava + pericardium

E. Superior vena cava + Pulmonary vein + pericardium

Answer: A. Pulmonary vein + Inferior vena cava + pericardium

Why: The oblique sinus is walled laterally by the pericardial reflections around the **pulmonary veins and IVC**, and posteriorly by pericardium over the esophagus.

60. During an open chest surgery, the surgeon threaded his index finger between the venous and arterial mesocardia, which of the following vessels will end up posterior to the surgeon's finger?

A. Inferior vena cava

B. Ascending aorta

C. Left subclavian vein

D. Superior vena cava

E. Pulmonary trunk

Answer: D. Superior vena cava

Why: Between the arterial and venous mesocardia is the transverse sinus; the arteries are in front and the **SVC** is behind the finger.

Surfaces, borders, apex and base

61. In beginning a coronary bypass procedure, a thoracic surgeon opens his patient's chest and pericardial sac, exposing the anterior surface of the heart. The anterior (sternocostal) surface of the heart is formed mainly by which of the following?

A. Right ventricle.

B. Left ventricle.

C. Right auricle.

D. Left auricle.

E. Left atrium.

Answer: A. Right ventricle.

Why: The anterior (sternocostal) surface is formed mainly by the **right ventricle**.

62. An 18-year-old male is brought to the emergency department after rear-ending a school bus that was stopped to off-load children. He was unrestrained and sustained blunt trauma over the sternum from the steering wheel. He has extensive bruising to the anterior thoracic wall. Which part of the heart would most likely be injured by the impact with the steering wheel?

A. Apex of left ventricle

B. Left ventricle

C. Right atrium

D. Right ventricle

Answer: D. Right ventricle

Why: The **right ventricle** forms most of the sternocostal surface, so steering-wheel impact on the sternum injures it most.

63. A patient sustains a stab wound just to the right of the sternum near the second intercostal space. Which heart chamber is most likely injured based on its anatomical location?

A. Left atrium

B. Left ventricle

C. Right atrium

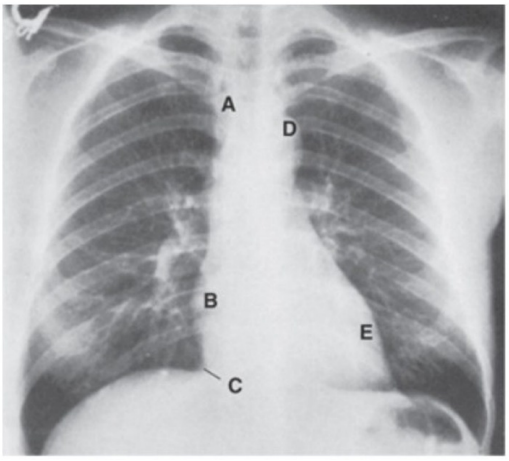
D. Right ventricle

E. Right atrium and left ventricle

Answer: C. Right atrium

Why: The **right atrium** forms the right border of the heart, the only chamber lying just right of the sternum.

64. Which of the labeled structures in the given PA radiograph of the chest indicates the right atrium?



A. A
B. B
C. C
D. D
E. E

Answer: B. B

Why: On a PA film the lower right heart border (**B**) is the **right atrium**; A is the SVC, D the aortic knob, E the left ventricle.

65. Which of the following associations is incorrect

A. right border of the heart—right atrium
B. diaphragmatic surface of the heart—mainly left ventricle
C. left border of the heart—mainly left atrium
D. superior border—right and left atria and auricles
E. anterior surface of the heart—mainly right ventricle

Answer: C. left border of the heart—mainly left atrium

Why: The left border is formed mainly by the **left ventricle** (slightly the left auricle), not the left atrium.

66. A 22-year-old medical student is reviewing a chest CT scan of a patient with blunt thoracic trauma. The heart is observed within the middle mediastinum, with its apex directed inferiorly, anteriorly, and to the left. Which of the following combinations correctly identifies the cardiac orientation, the chamber forming most of its anterior surface, and the structures forming its inferior border?

A. The heart is oriented with its base directed anteriorly; the left ventricle forms most of the anterior surface; the left atrium forms the inferior border.
B. The heart is oriented with its apex directed inferiorly, anteriorly, and to the left; the right ventricle forms most of the anterior surface; the right ventricle and right atrium form the inferior border.
C. The heart is oriented with its apex directed superiorly and to the right; the right atrium forms most of the anterior surface; the left ventricle forms the inferior border.
D. The heart is oriented with its base directed inferiorly; the right ventricle forms most of the posterior surface; the left atrium and left ventricle form the right border.
E. The heart is oriented with its apex directed posteriorly; the left atrium forms most of the anterior surface; the superior vena cava and pulmonary trunk form the inferior border.

Answer: B. The heart is oriented with its apex directed inferiorly, anteriorly, and to the left; the right ventricle forms most of the anterior surface; the right ventricle and right atrium form the inferior border.

Why: Apex points **inferiorly, anteriorly and left**; the **RV** forms most of the anterior surface. (Strictly, the inferior border is RV mainly plus a little LV.)

67. A 59-year-old man is admitted to the hospital with severe chest pain. During examination a slight rhythmic pulsation on the chest wall at the left fifth intercostal space is noted in the midclavicular line. What part of the heart is responsible for this pulsation?

A. Right atrium
B. Left atrium
C. Aortic arch
D. Apex of the heart
E. Tricuspid valve

Answer: D. Apex of the heart

Why: The **apex** (left ventricle) lies behind the **left 5th ICS at the midclavicular line**, producing the apex beat.

Heart wall and fibrous skeleton

68. A 4-year-old child is evaluated for recurrent respiratory infections and poor weight gain. On examination, a physician detects a systolic murmur at the left lower sternal border. Echocardiography reveals an opening in the muscular wall separating the two ventricles, with blood flowing from the left ventricle to the right ventricle. The physician explains that the heart contains two major septa that separate its chambers and prevent the mixing of blood under normal conditions. Which of the following correctly identifies the affected septum and distinguishes it from the other major cardiac septum?

A. The interatrial septum is affected; it separates the ventricles, while the interventricular septum separates the atria.

B. The interventricular septum is affected; it separates the right and left ventricles, while the interatrial septum separates the right and left atria.

C. The atrioventricular septum is affected; it separates the right and left ventricles, while the interventricular septum separates the right and left atria.

D. The interventricular septum is affected; it separates the right atrium from the right ventricle, while the interatrial septum separates the left atrium from the left ventricle.

E. The interatrial septum is affected; it separates the left atrium from the left ventricle, while the interventricular septum separates the right atrium from the right ventricle.

Answer: B. The interventricular septum is affected; it separates the right and left ventricles, while the interatrial septum separates the right and left atria.

Why: The **interventricular septum** separates the two ventricles; the **interatrial septum** separates the two atria.

Right atrium

69. A 24-year-old medical student examines a preserved human heart during an anatomy laboratory session. The instructor points to a chamber with a relatively thin muscular wall, prominent pectinate muscles in its auricle, and openings for the superior and inferior venae cavae and the coronary sinus. The instructor then asks the student to identify the chamber and explain how its structural features relate to its function. Which of the following best identifies this chamber and correctly describes its anatomical features?

A. Left atrium; it receives oxygenated blood from the pulmonary veins and contains prominent pectinate muscles throughout its entire chamber.

B. Right ventricle; it receives systemic venous blood directly from the venae cavae and contains the crista terminalis separating smooth and rough regions.

C. Left ventricle; it receives oxygenated blood from the pulmonary veins and contains the fossa ovalis on its interatrial septum.

D. Right atrium; it receives systemic venous blood, contains the fossa ovalis on its interatrial septum, and has a smooth posterior portion called the sinus venarum.

E. Right atrium; it receives oxygenated blood from the pulmonary veins and ejects blood into the aorta through the aortic valve.

Answer: D. Right atrium; it receives systemic venous blood, contains the fossa ovalis on its interatrial septum, and has a smooth posterior portion called the sinus venarum.

Why: Thin wall, pectinate auricle, SVC/IVC/coronary sinus openings = **right atrium**; it has the fossa ovalis and smooth **sinus venarum**.

70. What is the name of the structure that represents the foramen ovale (the closed version) and which chamber of the heart does it reside in?

A. Crista terminalis, left atrium

B. Fossa ovalis, right atrium

C. Pectinate muscles, right atrium

D. Trabeculae carneae, left ventricle

Answer: B. Fossa ovalis, right atrium

Why: The closed foramen ovale is the **fossa ovalis**, on the septal wall of the **right atrium**.

71. A 4-year-old with a small interventricular septal defect is undergoing surgical repair. To access the interventricular septum, a wide incision is made in the anterior wall of the right atrium to allow for the passage of instruments through the tricuspid valve into the right ventricle. Which structure could potentially be injured during the opening of the right atrium?

A. Crista terminalis

B. Trabeculae carneae

C. Conus arteriosus

D. Papillary muscle

E. Fossa ovalis

Answer: A. Crista terminalis

Why: The **crista terminalis** is a ridge in the RA wall (SVC to IVC) carrying conduction tissue; trabeculae, conus and papillary muscles are ventricular.

72. Which of the following structures does NOT drain into the right atrium?

A. SVC

B. IVC

C. Coronary sinus

D. Anterior cardiac vein

E. Pulmonary vein

Answer: E. Pulmonary vein

Why: Pulmonary veins drain into the **left atrium**; SVC, IVC, coronary sinus and anterior cardiac veins all open into the right atrium.

73. During an open heart surgery, a software malfunction causes the operating robot to misfire, making one of the surgical tools pierce through the inferior aspect of the interatrial septum. Which of the following structures could possibly have been damaged by this action?

A. SA node

B. Mitral valve

C. AV node

D. Pulmonary valve

E. Moderator band

Answer: C. AV node

Why: The **AV node** lies in the **inferior interatrial septum** near the coronary sinus opening; the SA node is high, at the SVC.

74. A radiologist is examining a series of contrast-enhanced CT scans of a patient's thorax in evaluating findings of a hypertrophied right heart. Which of the following structures is located in the pathologically enlarged right atrium?

A. Opening of the coronary sinus

B. Openings of the pulmonary veins

C. Septomarginal trabecula

D. Openings of the coronary arteries

E. Trabeculae carneae

Answer: A. Opening of the coronary sinus

Why: The **coronary sinus opens into the right atrium** between the IVC opening and the tricuspid orifice; pulmonary veins enter the left atrium.

Right ventricle and tricuspid valve

75. During ventricular systole, the atrioventricular valve cusps are prevented from being pushed backward into the atria. Which structures are directly responsible for maintaining this support?

A. Pectinate muscles and crista terminalis

B. Papillary muscles and chordae tendineae

C. Trabeculae carneae and semilunar sinuses

D. Fibrous trigones and coronary sinuses

Answer: B. Papillary muscles and chordae tendineae

Why: Papillary muscles contract and pull on **chordae tendineae**, holding AV cusps so they do not evert into the atria in systole.

Left atrium, left ventricle, mitral valve

76. Where is the bicuspid valve located?

A. Between the right ventricle and the aorta

B. Between the left ventricle and the aorta

C. Between the right ventricle and the right atria

D. Between the left ventricle and the left atria

Answer: D. Between the left ventricle and the left atria

Why: The **bicuspid (mitral)** valve lies between the **left atrium and left ventricle**.

77. During a cardiac anatomy laboratory, a first-year medical student examines a preserved heart. The instructor identifies a chamber that receives oxygenated blood from four pulmonary veins and communicates with the left ventricle through an atrioventricular valve. Most of the chamber's interior is smooth, but its auricle contains muscular ridges. An abnormal opening in the interatrial septum is also being discussed in relation to fetal circulation. Which of the following best identifies the chamber's anatomical features and their functional significance?

A. The left atrium receives oxygenated blood from the pulmonary veins, has a predominantly smooth-walled interior, contains pectinate muscles mainly in its auricle, and empties into the left ventricle through the mitral valve.

B. The left atrium receives deoxygenated blood from the venae cavae, has a smooth posterior wall called the sinus venarum, and empties into the left ventricle through the tricuspid valve.

C. The left atrium receives oxygenated blood from the pulmonary veins, contains the moderator band and prominent trabeculae carneae, and ejects blood directly into the aorta.

D. The left atrium receives oxygenated blood from the pulmonary veins, contains the crista terminalis separating smooth and rough regions, and empties into the left ventricle through the mitral valve.

E. The left atrium receives oxygenated blood from the pulmonary veins, has a predominantly smooth-walled interior, and ejects blood directly into the pulmonary trunk through the pulmonary valve.

Answer: A. The left atrium receives oxygenated blood from the pulmonary veins, has a predominantly smooth-walled interior, contains pectinate muscles mainly in its auricle, and empties into the left ventricle through the mitral valve.

Why: The LA receives the four pulmonary veins, is mostly **smooth-walled**, has pectinate muscles only in its **auricle**, and empties via the **mitral** valve.

78. A 68-year-old woman presents with progressive shortness of breath, fatigue, and difficulty breathing when lying flat. On cardiac auscultation, a high-pitched, blowing holosystolic murmur is heard loudest at the cardiac apex and radiates toward the left axilla. Echocardiography reveals that the mitral valve leaflets do not close completely during ventricular systole, allowing blood to flow backward into the left atrium. Which of the following best explains the functional impairment caused by this valve pathology and its likely clinical consequences?

A. Mitral stenosis reduces blood flow from the left atrium to the left ventricle during diastole, increasing left atrial pressure and potentially causing pulmonary congestion.

B. Mitral regurgitation allows blood to flow backward from the left ventricle into the left atrium during systole, increasing left atrial and pulmonary venous pressures and potentially causing pulmonary congestion and reduced forward cardiac output.

C. Aortic stenosis allows blood to flow backward from the aorta into the left ventricle during diastole, causing left ventricular volume overload and pulmonary edema.

D. Aortic regurgitation obstructs blood flow from the left ventricle into the aorta during systole, increasing left atrial pressure and causing a holosystolic murmur at the apex.

E. Tricuspid regurgitation allows blood to flow backward from the right atrium into the right ventricle during systole, increasing pulmonary venous pressure and causing left-sided heart failure as its primary manifestation.

Answer: B. Mitral regurgitation allows blood to flow backward from the left ventricle into the left atrium during systole, increasing left atrial and pulmonary venous pressures and potentially causing pulmonary congestion and reduced forward cardiac output.

Why: Incomplete mitral closure = **mitral regurgitation**: LV blood leaks into the LA in systole, raising LA and pulmonary venous pressure.

Aortic and semilunar valves

79. A 72-year-old man presents with progressive shortness of breath during exertion and episodes of dizziness. Cardiac auscultation reveals a harsh crescendo-decrescendo systolic murmur that is loudest at the right second intercostal space and radiates to the carotid arteries. Echocardiography shows thickened, calcified valve leaflets with restricted opening and an elevated pressure gradient between the left ventricle and the aorta during systole. Which of the following best explains the valve abnormality and the normal mechanism responsible for opening and closing this valve?

A. Mitral stenosis; the valve opens during ventricular systole to allow blood to flow from the left ventricle into the aorta.

B. Aortic stenosis; the valve opens during ventricular systole when left ventricular pressure exceeds aortic pressure and closes when aortic pressure exceeds left ventricular pressure.

C. Aortic regurgitation; the valve opens during ventricular diastole to allow blood to flow from the aorta into the left ventricle.

D. Pulmonary stenosis; the valve opens during ventricular diastole when right ventricular pressure exceeds pulmonary artery pressure.

E. Tricuspid regurgitation; the valve closes during ventricular systole when right atrial pressure exceeds right ventricular pressure.

Answer: B. Aortic stenosis; the valve opens during ventricular systole when left ventricular pressure exceeds aortic pressure and closes when aortic pressure exceeds left ventricular pressure.

Why: Aortic stenosis: the valve opens when **LV pressure exceeds aortic** pressure and closes when aortic pressure exceeds LV pressure.

Auscultation sites and chamber review

80. A 19-year-old patient is evaluated for a congenital cardiac abnormality. During an echocardiogram, a medical student observes the movement of a red blood cell returning from the systemic circulation. The instructor asks the student to identify the correct sequence of structures through which the red blood cell travels. Which of the following sequences correctly describes the normal path of blood through the heart and pulmonary circulation?

A. Superior and inferior venae cavae → right atrium → mitral valve → right ventricle → pulmonary valve → pulmonary trunk → lungs → pulmonary veins → left atrium → tricuspid valve → left ventricle → aortic valve → aorta.

B. Superior and inferior venae cavae → right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary trunk → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral valve → left ventricle → aortic valve → aorta.

C. Superior and inferior venae cavae → left atrium → tricuspid valve → left ventricle → pulmonary valve → pulmonary trunk → lungs → pulmonary veins → right atrium → mitral valve → right ventricle → aorta.

D. Superior and inferior venae cavae → right atrium → tricuspid valve → right ventricle → aortic valve → aorta → lungs → pulmonary arteries → pulmonary veins → left atrium → mitral valve → left ventricle → pulmonary trunk.

E. Superior and inferior venae cavae → right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary veins → lungs → pulmonary trunk → left atrium → mitral valve → left ventricle → aortic valve → aorta.

Answer: B. Superior and inferior venae cavae → right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary trunk → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral valve → left ventricle → aortic valve → aorta.

Why: Venae cavae → RA → **tricuspid** → RV → pulmonary valve → trunk → arteries → lungs → veins → LA → **mitral** → LV → aortic valve → aorta.

81. A 56-year-old woman undergoes a chest CT scan to evaluate a mediastinal mass. The mass compresses a large vessel entering the superior aspect of the right atrium, causing facial swelling and distension of the neck veins. During the imaging review, a medical student is asked to identify the great vessels based on their connections to the heart and the direction of blood flow. Which of the following correctly describes the anatomical relationships and functions of all these great vessels?

A. The superior vena cava enters the left atrium; the inferior vena cava enters the right atrium; the pulmonary trunk carries oxygenated blood to the lungs; the pulmonary veins return deoxygenated blood to the right ventricle; and the aorta arises from the left ventricle.

B. The superior vena cava enters the right ventricle; the inferior vena cava enters the right atrium; the pulmonary trunk arises from the left ventricle; the pulmonary veins enter the right atrium; and the aorta arises from the right ventricle.

C. The superior and inferior venae cavae enter the right atrium; the pulmonary trunk arises from the right ventricle and carries deoxygenated blood toward the lungs; the pulmonary veins return oxygenated blood to the left atrium; and the aorta arises from the left ventricle.

D. The superior and inferior venae cavae enter the right atrium; the pulmonary trunk arises from the left ventricle; the pulmonary veins return oxygenated blood to the left atrium; and the aorta arises from the right ventricle.

E. The superior vena cava enters the right atrium; the inferior vena cava enters the left atrium; the pulmonary trunk carries deoxygenated blood to the lungs; the pulmonary veins return oxygenated blood to the left atrium; and the aorta arises from the left ventricle.

Answer: C. The superior and inferior venae cavae enter the right atrium; the pulmonary trunk arises from the right ventricle and carries deoxygenated blood toward the lungs; the pulmonary veins return oxygenated blood to the left atrium; and the aorta arises from the left ventricle.

Why: Both venae cavae → **RA**; pulmonary trunk from **RV** (deoxygenated); pulmonary veins → **LA** (oxygenated); aorta from **LV**.

82. A 92-year-old male is admitted to the hospital due to excruciating anginal pains. During the consultation, the physician auscultates the patient's heart and notices the first heart sound is much louder than the second. The closing of which of the following valves is indicated by the first heart sound?

A. Tricuspid and Aortic

B. Bicuspid and Pulmonary

C. Pulmonary and Aortic

D. Bicuspid and Tricuspid

E. Aortic and Bicuspid

Answer: D. Bicuspid and Tricuspid

Why: S1 ("lub") is closure of the **AV valves**: tricuspid and bicuspid (mitral).

83. A 72-year-old female is brought to the emergency department due to substernal, pressure-like pain that is radiating down her left arm. ECG shows a myocardial infarction and a right bundle branch block. Auscultation of the heart shows a loud second heart sound. Closure of which valves are responsible for this second sound?

A. Tricuspid and bicuspid valves

B. Aortic and tricuspid valves

C. Mitral and pulmonary valves

D. Tricuspid and pulmonary valves

E. Aortic and pulmonary valves

Answer: E. Aortic and pulmonary valves

Why: S2 is closure of the **aortic and pulmonary** (semilunar) valves.

84. A 45-year-old female goes to her physician for a routine examination. Vital signs are within normal limits, and physical examination reveals a prominent S1 heart sound located in the left 5th intercostal space in the mid-clavicular line. Which valve is best auscultated from this location?

A. Pulmonary valve

B. Aortic valve

C. Tricuspid valve

D. Mitral valve

Answer: D. Mitral valve

Why: The left 5th ICS at the midclavicular line is the **mitral** area.

85. A 65-year-old man presents with progressive exertional dyspnea. During cardiac auscultation, a harsh systolic murmur is heard most prominently at the right second intercostal space, adjacent to the sternum. The murmur radiates toward the carotid arteries. The examiner also wants to assess normal S1 and S2 heart sounds and compare their intensity at different auscultation sites. Which of the following best integrates the findings with the appropriate clinical auscultation landmarks?

A. The murmur suggests mitral regurgitation; it should be loudest at the apex and radiate toward the left axilla, while S1 is generally louder at the base.

B. The murmur suggests aortic stenosis; it is heard at the right second intercostal space and may radiate to the neck, while S2 is generally louder at the apex.

C. The murmur suggests aortic stenosis; it is best heard at the right second intercostal space and may radiate to the neck, while S1 is generally louder at the apex and S2 at the base.

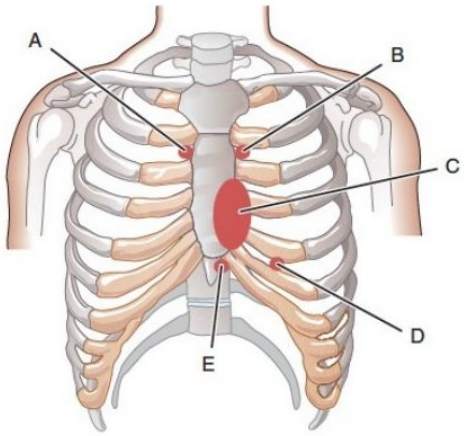
D. The murmur suggests pulmonary stenosis; it is best heard at the left second intercostal space, while S1 and S2 are generally equally intense at the apex and base.

E. The murmur suggests tricuspid regurgitation; it is best heard at the lower left sternal border and increases with inspiration, while S1 is generally louder at the base.

Answer: C. The murmur suggests aortic stenosis; it is best heard at the right second intercostal space and may radiate to the neck, while S1 is generally louder at the apex and S2 at the base.

Why: Right 2nd ICS murmur radiating to the carotids = **aortic stenosis**; **S1 is louder at the apex**, **S2 at the base**.

86. A 50-year old male notices he has an increased breathing rate even while just sitting down without having exerted himself. He goes to his physician for a physical exam which reveals a pulmonary valve defect. Which of the following locations is the proper place to auscultate the pulmonary valve?



A. A

B. B

C. C

D. D

E. E

Answer: B. B

Why: The pulmonary area is the **left 2nd ICS** just lateral to the sternum (label B).

Coronary arteries and dominance

87. What is the first branch given off by the Right coronary artery?

A. Right Marginal branch

B. AV (atrioventricular) nodal branch

C. SA nodal branch

D. Posterior interventricular branch

Answer: C. SA nodal branch

Why: The **SA nodal branch** leaves the RCA near its origin; the marginal, AV nodal and PDA come later along its course.

88. A 65-year-old male has severe chest pain, which he describes as crushing and radiating down his left arm. An ECG shows a myocardial infarction and ventricular arrhythmia. Coronary angiography show a blockage of the right coronary artery distal to the origin of the right marginal artery (posteriorly in the atrioventricular sulcus) in a right coronary dominant circulation. Which structure would most likely be affected by this blockage?

A. AV node

B. Left atrium

C. SA node

D. Lateral wall of left ventricle

E. Anterior interventricular septum

Answer: A. AV node

Why: Past the right marginal, the RCA still gives the **AV nodal branch at the crux**, so a block there starves the AV node.

89. A patient comes to your office with severe chest pain. You immediately order a scan. Looking at the scan you realize the patient's right ventricle is severely ischemic. Which of the following blood vessels is the prime supplier of the right ventricle?

A. Right atrial branch

B. Left IV septal branch

C. Right marginal branch

D. Circumflex branch

Answer: C. Right marginal branch

Why: The **right marginal branch** of the RCA supplies most of the right ventricle.

90. The right coronary artery supplies all of the following structures except:

A. The SA node (in most individuals)

B. The AV node (in most individuals)

C. The right atrium

D. The lateral wall of the left ventricle

E. The diaphragmatic surface of the right ventricle

Answer: D. The lateral wall of the left ventricle

Why: The **lateral wall of the LV** is circumflex territory; the RCA supplies the SA/AV nodes, right atrium and RV.

91. An 83 year old man with a typical coronary circulation has been suffering from an embolism of the circumflex branch of the left coronary artery. This condition would result in ischemia of which of the following areas of the heart?

A. Anterior part of LV

B. Anterior interventricular region

C. Posterior LV

D. Posterior interventricular region

E. Anterior part of RV

Answer: C. Posterior LV

Why: The **circumflex** wraps to the back of the left side, so its occlusion leaves the **posterior (posterolateral) LV** ischemic.

92. A 55-year-old patient is to undergo a coronary bypass operation. The artery of primary concern is the vessel that supplies much of the left ventricle and the right and left bundle branches of the cardiac conduction system. Which artery is the surgeon most concerned with?

A. Right marginal

B. Anterior interventricular

C. Circumflex

D. Artery to the sinoatrial (SA) node

E. Posterior interventricular

Answer: B. Anterior interventricular

Why: The **anterior interventricular artery** supplies much of the LV and, by septal branches, the bundle branches in the anterior septum.

93. A 60 year old female is scheduled for an angiogram. The scan shows a narrowing of her anterior interventricular artery (LAD) close to its origin. Which of the following arteries would be able to provide blood to the occluded part of the anterior interventricular artery if the anastomosis was large enough?

A. Circumflex

B. Posterior interventricular

C. Right coronary

D. Left coronary

E. Left marginal

Answer: B. Posterior interventricular

Why: The LAD **anastomoses with the posterior interventricular artery** at the apex, so the PDA can backfill an occluded LAD.

94. A 64-year-old man with a history of hypertension and hyperlipidemia presents with episodes of dizziness and near-syncope. An ECG reveals sinus bradycardia with intermittent pauses. Coronary angiography demonstrates an atherosclerotic narrowing of the artery supplying the sinoatrial (SA) node. The cardiologist explains that ischemia of the SA node can impair the initiation of electrical impulses and disrupt the normal heart rate. Which of the following best explains the anatomical origin of the affected artery and the likely consequence of its atherosclerotic narrowing?

A. The SA nodal artery usually arises from the left anterior descending artery, and its obstruction primarily delays conduction through the atrioventricular node.

B. The SA nodal artery commonly arises from the circumflex artery only, and its obstruction primarily causes complete atrioventricular block without affecting the SA node.

C. The SA nodal artery always arises from the RCA, and its obstruction primarily causes ventricular fibrillation by interrupting the Purkinje fibers.

D. The SA nodal artery usually arises from the posterior descending artery, and its obstruction primarily blocks conduction through the left bundle branch.

E. The SA nodal artery commonly arises from the RCA or the circumflex artery, and its obstruction may impair pacemaker activity, causing sinus bradycardia or sinoatrial dysfunction.

Answer: E. The SA nodal artery commonly arises from the RCA or the circumflex artery, and its obstruction may impair pacemaker activity, causing sinus bradycardia or sinoatrial dysfunction.

Why: The SA nodal artery comes from the **RCA (~60%) or circumflex (~40%)**; narrowing it starves the pacemaker, giving sinus bradycardia and pauses.

95. A 25-year-old patient undergoes repair of an atrial septal defect. After surgery, the rhythm becomes markedly erratic, and testing suggests that blood supply to both the SA node and AV node was disrupted. Which artery was most likely injured?

A. Left anterior descending (LAD) artery

B. Circumflex artery

C. Right coronary artery (RCA)

D. Left marginal artery

E. Posterior interventricular artery (PDA)

Answer: C. Right coronary artery (RCA)

Why: The **RCA** usually supplies both the SA node (60%) and the AV node (80%), so one RCA injury disturbs both.

96. A 61-year-old man undergoes coronary angiography after experiencing chest pain on exertion. The cardiologist also identifies a significant stenosis in the proximal RCA. A medical student is asked to explain how the origin of the PDA determines coronary dominance and why this anatomical variation is clinically important when predicting myocardial ischemia. Which of the following statements best explains this patient's coronary anatomy and its clinical significance?

A. The patient has left-dominant circulation because the PDA supplies the posterior interventricular septum, and stenosis of the RCA primarily threatens the anterior wall of the left ventricle.

B. The patient has left-dominant circulation because the RCA supplies the right atrium, and proximal RCA stenosis primarily compromises blood flow to the lateral wall of the left ventricle.

C. The patient has codominant circulation because both coronary arteries supply the myocardium, and coronary dominance is determined by which artery gives rise to the sinoatrial nodal artery.

D. The patient has right-dominant circulation because the PDA arises from the RCA, and proximal RCA stenosis may compromise blood supply to the inferior myocardium and, depending on the anatomy, the atrioventricular node.

E. The patient has right-dominant circulation because the PDA arises from the RCA, but proximal RCA stenosis cannot affect the atrioventricular node because its blood supply is exclusively from the left coronary artery.

Answer: D. The patient has right-dominant circulation because the PDA arises from the RCA, and proximal RCA stenosis may compromise blood supply to the inferior myocardium and, depending on the anatomy, the atrioventricular node.

Why: The **PDA arises from the RCA** (right dominant), so proximal RCA stenosis threatens the inferior wall and often the AV node.

97. A 58-year-old man is undergoing coronary angiography following a myocardial infarction involving the posterior wall of the left ventricle. The imaging reveals that the posterior descending artery (PDA) arises from the circumflex branch of the left coronary artery. Which of the following best describes this patient’s coronary circulation pattern?

A. Right dominant

B. Left dominant

C. Codominant

D. Anomalous right coronary origin

E. Balanced Supply

Answer: B. Left dominant

Why: A PDA from the **circumflex** is left dominance.

98. Sometimes when people have a “boring” job, they need to put up pictures of things they’ve bought or places they’ve traveled to, to say, “Hey check out my FaceBook profile, I’m interesting”. Anyway, this “competition” absorbs most of Greg’s life, who sits behind a cubicle and crunches numbers. As he continues his sitting, his PDA is clogging up every year until he shows up to the Cath Lab. The attending asks you, what are the percentages of people who are Right Dominant, Left Dominant or Co-Dominant?

A. 70%, 20%, 10%

B. 20%, 10%, 70%

C. 70%, 10%, 20%

D. 20%, 70%, 10%

Answer: C. 70%, 10%, 20%

Why: Right **70%**, left **10%**, co-dominant **20%**.

99. During an autopsy, a coroner finds a blockage in the left coronary artery at its origin. While analyzing the heart, he notices that the entire ventricular septum was ischemic indicating the patient was left heart dominant. This indicates that which of the following branches came off of the left coronary artery?

A. Right marginal

B. Left marginal

C. Right coronary

D. Posterior interventricular

E. SA nodal

Answer: D. Posterior interventricular

Why: In left dominance the LCA also gives the **posterior interventricular artery**, so an LCA block takes the whole septum.

100. A 63-year-old male goes to the emergency department after having chest pain with exertion. Coronary angiography shows a left dominant heart and the left coronary artery to be 70-80% occluded proximal to its bifurcation into the circumflex and anterior interventricular arteries. Which explanation is the most likely for a poor prognosis if this patient has no surgical intervention?

A. The blood supply of the SA node is insufficient

B. All coronary artery branches are end arteries – there is no chance for an anastomosis

C. Development of effective collateral circulation between the posterior (inferior) and anterior interventricular aa. will not be possible

D. The blood supply of the AV node is insufficient

Answer: C. Development of effective collateral circulation between the posterior (inferior) and anterior interventricular aa. will not be possible

Why: With left dominance the **PDA comes from the left system**, so no PIV-to-AIV collateral can rescue a proximal LCA block.

Cardiac veins and coronary occlusion

101. Which cardiac vein travels through the anterior interventricular sulcus alongside the left anterior descending (LAD) artery?

A. Middle cardiac vein

B. Small cardiac vein

C. Great cardiac vein

D. Anterior cardiac vein

E. Posterior vein of the left ventricle

Answer: C. Great cardiac vein

Why: The **great cardiac vein** runs with the LAD in the anterior interventricular sulcus.

102. A 48-year- old male patient is admitted with chronic angina. Coronary angiography reveals nearly total blockage of the circumflex artery near its origin from the left coronary artery. When this artery is exposed to perform a bypass surgery, what accompanying vein must be protected from injury?

A. Middle cardiac

B. Great cardiac

C. Small cardiac

D. Anterior cardiac

E. Posterior cardiac

Answer: B. Great cardiac

Why: After leaving the LAD, the **great cardiac vein** turns into the coronary sulcus alongside the **circumflex** to join the coronary sinus.

103. A 59-year-old female is scheduled for a bypass procedure due to the blockage found in her right posterior interventricular artery. Which accompanying vein must be protected during the bypass procedure?

A. Great cardiac

B. Middle cardiac

C. Small cardiac

D. Coronary sinus

E. Oblique

Answer: B. Middle cardiac

Why: The posterior interventricular artery is accompanied by the **middle cardiac vein**.

104. 41 A 42 year old male is brought in for a cardiac arrest, the surgeon has to bypass the right marginal artery, what neighboring vein would be affected?

A. Small cardiac vein

B. Anterior cardiac vein

C. Great cardiac vein

D. Inferior cardiac vein

E. Middle cardiac vein

Answer: A. Small cardiac vein

Why: The **small cardiac vein** runs with the right marginal artery along the acute margin and then with the RCA.

105. The Coronary sinus receives all venous drainage EXCEPT:

A. Small cardiac vein

B. Oblique vein of left atrium

C. Middle cardiac vein

D. Anterior cardiac veins

Answer: D. Anterior cardiac veins

Why: The **anterior cardiac veins** drain the anterior RV straight into the right atrium, bypassing the coronary sinus.

106. Atherosclerotic occlusion of the LAD most likely produces necrosis in which anatomical region of the heart?

A. Inferior wall of left ventricle

B. Posterior wall of left ventricle

C. Anterior wall of left ventricle and anterior 2/3 of septum

D. Lateral wall of left ventricle

E. Right ventricle

Answer: C. Anterior wall of left ventricle and anterior 2/3 of septum

Why: The LAD supplies the **anterior LV wall, anterior 2/3 of the septum** and apex.

107. A 52-year-old man collapses while jogging and is brought to the emergency department. Emergent coronary angiography reveals a proximal occlusion of a major coronary artery (see image). Occlusion of which of the following arteries would produce this angiographic finding and is commonly referred to as the “widowmaker”

A. Right coronary artery (RCA)

B. Left circumflex artery (LCx)

C. Left anterior descending artery (LAD)

D. Posterior descending artery (PDA)

E. Left marginal branch

Answer: C. Left anterior descending artery (LAD)

Why: The **LAD** is called the widowmaker; proximal occlusion infarcts the anterior LV, septum and apex.

108. A 64-year-old man develops an acute anterior myocardial infarction. Shortly afterward, his ECG shows a new conduction defect affecting activation of both ventricles. Ischemia of which structure is the best anatomic explanation?

A. SA node in the superior right atrial wall

B. AV node in the inferior interatrial septum

C. AV bundle and proximal bundle branches in the interventricular septum

D. Terminal Purkinje fibers confined to the right ventricular free wall

E. Right vagal cardiac branches

Answer: C. AV bundle and proximal bundle branches in the interventricular septum

Why: The LAD feeds the anterior 2/3 of the septum, where the **AV bundle and bundle branches** run, so anterior MI blocks both-ventricle activation.

109. An inferior myocardial infarction is followed by AV nodal block in a right-dominant patient. Which artery was most likely involved?

A. LAD

B. RCA

C. Left marginal

D. Pulmonary trunk

E. Great cardiac vein

Answer: B. RCA

Why: The **RCA** supplies the inferior wall and gives the **AV nodal branch** (about 80% of people), so **inferior MI + AV block** points to the RCA.

Conduction system and ECG

110. A 46-year-old man undergoes electrophysiologic mapping for palpitations. The earliest spontaneous depolarization is recorded from a small nodal structure located in the superior wall of the right atrium near the entrance of the superior vena cava, along the crista terminalis. Which structure is being identified?

A. Atrioventricular (AV) node

B. Bundle of His

C. Purkinje fibers

D. Sinoatrial (SA) node

E. Right bundle branch

Answer: D. Sinoatrial (SA) node

Why: The **SA node** sits at the upper end of the crista terminalis near the SVC opening and fires first.

111. Following a myocardial infarction, an irregular cardiac rhythm is detected, and coronary angiography shows a blocked anterior interventricular artery. A stent is placed in the artery, and a cardiac pacemaker is also placed in the heart. Which structure is essentially replaced by the insertion of a pacemaker?

A. AV node

B. SA node

C. Bundle of His

D. Purkinje fibers

Answer: B. SA node

Why: An artificial pacemaker takes over the rate-setting job of the **SA node**.

112. During surgical closure of a ventricular septal defect in the membranous part of the septum, which conduction structure is most at risk?

A. SA node

B. AV bundle (of His)

C. Purkinje fibers

D. Right bundle branch in the septomarginal trabecula (moderator band)

E. Internodal pathways of the right atrium

Answer: B. AV bundle (of His)

Why: The **AV bundle (of His)** runs from the AV node along the **interventricular septum** (membranous part) before splitting at the top of the muscular septum, so membranous VSD repair risks heart block. The moderator band carries the right bundle branch far lower, to the anterior papillary muscle.

113. During repair of a congenital ventricular defect, a surgeon accidentally damages the septomarginal trabecula. Postoperatively, conduction to the anterior wall of the right ventricle is impaired. Which structure was most likely interrupted?

A. Left bundle branch

B. Right bundle branch

C. AV node

D. SA node

E. Left vagal cardiac branch

Answer: B. Right bundle branch

Why: The **septomarginal trabecula (moderator band)** carries the **right bundle branch** to the anterior papillary muscle and anterior RV wall.

114. A patient has a prolonged PR interval but normal QRS complexes. Which portion of the conduction system is most closely associated with the normal physiological delay represented by the PR interval?

A. Purkinje fibers

B. AV node

C. Ventricular myocardium

D. Left bundle branch only

E. SA nodal artery

Answer: B. AV node

Why: The **AV node** holds the impulse back so the atria finish emptying; this **AV delay** makes up most of the **PR interval**.

Innervation, procedures and clinical integration

115. A Fifty-five-year-old man presents with episodes of dizziness and a slow heart rate. An ECG reveals sinus bradycardia. Which of the following best describes the cardiac conduction pathway and the autonomic mechanism responsible for this decrease in heart rate?

A. The impulse travels from the SA node to the AV node, bundle of His, bundle branches, and Purkinje fibers; sympathetic preganglionic neurons originate in the upper thoracic spinal cord and synapse in cervical or thoracic sympathetic ganglia, whereas parasympathetic preganglionic neurons originate in the medulla and synapse in ganglia near or within the heart.

B. The impulse travels from the AV node to the SA node, Purkinje fibers, bundle branches, and bundle of His; sympathetic preganglionic neurons originate in the medulla and synapse within the cardiac wall, whereas parasympathetic preganglionic neurons originate in the thoracic spinal cord.

C. The impulse travels from the SA node directly to the Purkinje fibers before reaching the AV node; sympathetic nerves decrease heart rate through the vagus nerve, whereas parasympathetic nerves increase heart rate through the sympathetic chain.

D. The impulse travels from the SA node to the AV node and then directly to the ventricular myocardium without passing through the bundle of His; both sympathetic and parasympathetic preganglionic neurons synapse in the cervical sympathetic ganglia.

E. The impulse travels from the SA node to the AV node, bundle of His, bundle branches, and Purkinje fibers; parasympathetic preganglionic neurons originate in the upper thoracic spinal cord and synapse in the sympathetic chain, whereas sympathetic preganglionic neurons originate in the medulla and synapse near the heart.

Answer: A. The impulse travels from the SA node to the AV node, bundle of His, bundle branches, and Purkinje fibers; sympathetic preganglionic neurons originate in the upper thoracic spinal cord and synapse in cervical or thoracic sympathetic ganglia, whereas parasympathetic preganglionic neurons originate in the medulla and synapse in ganglia near or within the heart.

Why: Impulse runs **SA, AV, His, branches, Purkinje**; sympathetic preganglionics are upper thoracic cord, parasympathetic (vagal) preganglionics are medullary and synapse in cardiac ganglia.

116. A patient undergoes a procedure that interrupts preganglionic sympathetic input to the heart while vagal input remains intact. Which combination best describes the affected pathway and the expected effect?

A. C3–C5 → phrenic nerve; decreased heart rate

B. T1–T4 → sympathetic chain/cardiac nerves; decreased heart rate

C. Thoracic splanchnic nerves to abdominal viscera; increased heart rate

D. T10–L2 → lumbar splanchnic nerves; decreased heart rate

E. Medulla → vagus nerve; increased heart rate

Answer: B. T1–T4 → sympathetic chain/cardiac nerves; decreased heart rate

Why: Cardiac sympathetics come from **T1-T4** via the chain and cardiac nerves; removing them leaves the vagus unopposed, so the rate drops.

117. A 45 year old man presents to the ER with Shortness Of breath, light-headedness, tightness in his chest, and pain that radiates down his left arm. What nerve roots refer pain to his arm?

A. C2

B. C3

C. T1

D. T4

Answer: C. T1

Why: Cardiac pain afferents enter **T1-T2** (up to T4), so pain is referred to the medial arm along the T1-T2 dermatomes.

118. A 35-year-old man is brought to the emergency department after sustaining a stab wound to the chest. He is hypotensive, tachycardic, and has distended neck veins. Heart sounds are muffled, and bedside echocardiography reveals a large pericardial effusion with right ventricular diastolic collapse, suggesting cardiac tamponade. The emergency physician decides that urgent pericardiocentesis is required. The procedure must be performed through an access route that minimizes the risk of injury to the myocardium, lungs, and major blood vessels. Which of the following best describes an appropriate approach and its anatomical rationale?

A. Insert the needle through the right second intercostal space immediately adjacent to the sternum, because this provides direct access to the right atrium while avoiding the lungs and internal thoracic vessels.

B. Use an ultrasound-guided subxiphoid approach, directing the needle toward the pericardial fluid collection while avoiding the liver, myocardium, and other adjacent structures.

C. Insert the needle through the left fifth intercostal space at the midclavicular line without imaging, because the apex of the heart is the safest point for accessing the pericardial cavity.

D. Insert the needle through the third intercostal space at the left sternal border, because the internal thoracic vessels are located farthest from the sternum at this level.

E. Insert the needle through the xiphosternal junction directly into the right ventricle, because the ventricular cavity communicates with the pericardial cavity during cardiac tamponade.

Answer: B. Use an ultrasound-guided subxiphoid approach, directing the needle toward the pericardial fluid collection while avoiding the liver, myocardium, and other adjacent structures.

Why: An **ultrasound-guided subxiphoid** approach reaches the fluid while avoiding pleura, lungs and coronary/internal thoracic vessels.

119. A 65-year-old patient is brought to the emergency department because of severe chest pain. Examination reveals muffled heart sounds and distended jugular veins. Lab results show elevated troponins confirming myocardial infarction. Cardiac tamponade is suspected and an emergency pericardiocentesis is ordered. Where should the physician insert the needle to relieve the cardiac tamponade?

A. Triangle of Auscultation

B. Right 7th intercostal space in the mid-axillary line

C. Left 5th intercostal space at the sternal border

D. Left 6th intercostal space in the mid-clavicular line

E. Right 3rd intercostal space 1 inch lateral to the sternum

Answer: C. Left 5th intercostal space at the sternal border

Why: Besides subxiphoid, the needle can enter the **left 5th ICS at the sternal border**, where the pleura is pulled aside at the cardiac notch.

120. During tube thoracostomy, the operator enters the safe triangle. Where should the incision track relative to the rib to reduce injury to the intercostal neurovascular bundle?

A. Immediately below lower rib margin

B. Immediately above upper border of rib

C. Through the costal groove

D. Through the sternum

E. Through costal cartilage at midline

Answer: B. Immediately above upper border of rib

Why: The intercostal vein, artery and nerve run in the **costal groove** under the **lower border** of each rib, so the tube goes **just above the upper border** of the rib below.

Embryology of the cardiovascular system

Origins of the cardiovascular system

121. Which of the following embryonic tissues gives rise to the primordium of the heart during the formation of the primitive heart tube?

A. Paraxial mesoderm

B. Intermediate mesoderm

C. Splanchnic mesoderm

D. Ectoderm

E. Endoderm

Answer: C. Splanchnic mesoderm

Why: The heart primordium comes from the **splanchnic mesoderm** of the cranial lateral plate.

122. At week 4 of gestation, the primitive heart begins pumping blood through vessels even though the heart is still a simple tube that has not yet undergone septation. This early functional circulation is best explained by which principle of embryonic development?

A. Organ systems must be fully differentiated before they can function

B. Function often begins before an organ reaches its final structural form

C. The heart does not begin functioning until the fetal period

D. Blood flow at this stage is purely maternal and passes through the embryo passively

E. The primitive heart tube pumps lymph, not blood

Answer: B. Function often begins before an organ reaches its final structural form

Why: The tube heart pumps by week 4, long before septation: embryonic organs often **function before they reach final form**.

Embryonic veins and their fates

123. During the fourth week of development, three paired venous systems return blood toward the developing heart. Which pairing correctly identifies one of these vessels and the region from which it returns blood?

A. Vitelline vein — body of embryo

B. Vitelline vein — chorion

C. Common cardinal vein — umbilical vesicle

D. Umbilical vein — chorion

E. Umbilical vein — body of embryo

Answer: D. Umbilical vein — chorion

Why: Umbilical vein — chorion is correct; vitelline veins drain the umbilical vesicle and common cardinal veins drain the embryo body.

124. The portal vein is derived from which of the following?

A. Vitelline veins

B. Umbilical veins

C. Anterior cardinal veins

D. Posterior cardinal veins

E. Subcardinal veins

Answer: A. Vitelline veins

Why: The portal vein forms from the **vitelline veins** (mainly the right one) around the duodenum.

125. During the 7th week of development, a shunt develops between the umbilical vein and the heart. This shunt is

A. hepatic portal vein
B. vitelline veins
C. ductus venosus
D. ductus arteriosus
E. ductus umbilicus

Answer: C. ductus venosus

Why: The **ductus venosus** develops in the liver as a shunt from the umbilical vein straight to the IVC and heart.

126. Which umbilical vein persists prenatally, and how do vitelline remodeling and the ductus venosus affect its route?

A. Right umbilical persists; cardinal veins form portal channels; ductus venosus bypasses lungs
B. Both umbilical veins persist; vitelline veins form pulmonary veins; ductus venosus bypasses right ventricle
C. Left umbilical persists; vitelline veins form SVC; ductus venosus connects right and left atria
D. Left umbilical persists; vitelline veins help form portal and hepatic channels; ductus venosus bypasses much of liver

Answer: D. Left umbilical persists; vitelline veins help form portal and hepatic channels; ductus venosus bypasses much of liver

Why: The **left umbilical vein persists**; vitelline veins form **portal and hepatic** channels; the **ductus venosus bypasses much of the liver**.

127. During remodeling of the cardinal venous system, the right and left anterior cardinal veins develop an anastomosis that redirects blood from left to right. Which adult vessel develops from this shunt?

A. Superior vena cava
B. Inferior vena cava
C. Coronary sinus
D. Azygos vein
E. Left brachiocephalic vein

Answer: E. Left brachiocephalic vein

Why: The week-8 anastomosis carrying blood left → right between the anterior cardinals becomes the **left brachiocephalic vein**.

128. Which embryonic vessels combine to form the superior vena cava?

A. Right anterior cardinal and right common cardinal veins
B. Right anterior cardinal and right subcardinal veins
C. Right common cardinal and right supracardinal veins
D. Right vitelline and right common cardinal veins
E. Right subcardinal and right supracardinal veins

Answer: A. Right anterior cardinal and right common cardinal veins

Why: SVC = **right anterior cardinal + right common cardinal** veins.

129. The renal veins are derived from which of the following?

A. Vitelline veins
B. Umbilical veins
C. Anterior cardinal veins
D. Posterior cardinal veins
E. Subcardinal veins

Answer: E. Subcardinal veins

Why: The **subcardinal veins** form the stem of the left renal vein plus suprarenal and gonadal veins.

130. A developmental abnormality selectively affects the supracardinal veins. Which adult venous structures would be most directly affected?

A. SVC and left brachiocephalic vein
B. Hepatic portal vein and hepatic IVC
C. Azygos/hemiazygos system and part of the IVC
D. Pulmonary veins and coronary sinus
E. Left brachiocephalic and renal veins

Answer: C. Azygos/hemiazygos system and part of the IVC

Why: Supracardinals form the **azygos/hemiazygos** system cranially and the right one forms the postrenal IVC caudally.

Development of the inferior vena cava

131. Which embryonic venous system give rise to the hepatic segment of the inferior vena cava?

A. Left anterior cardinal vein
B. Right vitelline vein
C. Right subcardinal vein
D. Right supracardinal vein

Answer: B. Right vitelline vein

Why: The hepatic segment of the IVC comes from the proximal **right vitelline vein**.

132. A student is tracing the embryologic origin of the inferior vena cava. Which pairing is correct?

A. Hepatic segment — right supracardinal vein
B. Prerenal segment — subcardinal vein
C. Renal segment — right vitelline vein
D. Postrenal segment — left supracardinal vein
E. Hepatic segment — anterior cardinal vein

Answer: B. Prerenal segment — subcardinal vein

Why: IVC segments: hepatic = right vitelline, **prerenal = subcardinal**, renal = sub–supracardinal anastomosis, postrenal = right supracardinal.

Pharyngeal arch arteries and derivatives

133. Which structures arise from the right and left horns of the aortic sac?

- **A.** Right horn → right pulmonary artery; left horn → ductus arteriosus
- **B.** Right horn → right subclavian artery in its entirety; left horn → left subclavian artery in its entirety
- **C.** Right horn → SVC; left horn → coronary sinus
- **D.** Right horn → brachiocephalic trunk; left horn → proximal aortic arch

Answer: D. Right horn → brachiocephalic trunk; left horn → proximal aortic arch

Why: Right horn of the aortic sac → **brachiocephalic trunk**; **left horn → proximal aortic arch**.

134. Which of the following pairs of fetal aortic arches and its adult remnant is incorrect?

- **A.** 1st arch--- Maxillary A.
- **B.** 6th arch--- Left side> Left pulmonary A & ductus arteriosus Right>side Right pulmonary A
- **C.** 2nd arch-- Common carotid & first part of the internal carotid A's
- **D.** 4th arch-- Left side >Arch of the aorta between the left common carotid & left subclavian arteries Right side > Right subclavian artery (proximal portion)

Answer: C. 2nd arch-- Common carotid & first part of the internal carotid A's

Why: The **2nd arch** gives the stapedial artery; the common carotid and proximal internal carotid come from the 3rd.

135. A 3-day old neonate presents with constant flushing of the right side of his face. A carotid pulse check reveals a very strong right carotid pulse yet an extremely weak left carotid pulse. An arteriogram confirms the malformation of his left common carotid artery. Which of the following arterial arches gives rise to the common carotid arteries?

- **A.** 1st aortic arch
- **B.** 2nd aortic arch
- **C.** 3rd aortic arch
- **D.** 4th aortic arch
- **E.** 6th aortic arch

Answer: C. 3rd aortic arch

Why: The common carotids come from the proximal **3rd arch** arteries.

136. A developmental abnormality selectively disrupts the third pharyngeal arch artery. Which adult vessels would be most directly affected?

- **A.** Pulmonary arteries
- **B.** Common and internal carotid arteries
- **C.** Aortic arch and right subclavian artery
- **D.** Maxillary and stapedial arteries
- **E.** Ductus arteriosus and pulmonary arteries

Answer: B. Common and internal carotid arteries

Why: 3rd arch: proximal → **common carotid**, distal → **internal carotid**.

137. A 12-year-old boy undergoes a thorough workup for progressive, asymmetric upper extremity hypertension and weak femoral pulses. A diagnosis of coarctation of the aorta (postductal type) is confirmed via magnetic resonance angiography. The constricted segment of the aortic arch is located just distal to the origin of the left subclavian artery. An abnormality in the embryonic remodeling of which pharyngeal arch derived vessel is directly tied to the development of this segment of the definitive aorta?

- **A.** Third pharyngeal arch artery
- **B.** Left fourth pharyngeal arch artery
- **C.** Right fourth pharyngeal arch artery
- **D.** Left sixth pharyngeal arch artery
- **E.** Maxillary artery of the first pharyngeal arch

Answer: B. Left fourth pharyngeal arch artery

Why: The segment of aortic arch around the left subclavian comes from the **left 4th** arch artery (with left dorsal aorta).

138. The fourth pharyngeal arch arteries have different adult derivatives on the right and left. Which pairing is correct?

- **A.** Left → pulmonary artery; Right → ductus arteriosus
- **B.** Left → common carotid; Right → internal carotid
- **C.** Left → arch of aorta; Right → proximal right subclavian artery
- **D.** Left → descending aorta; Right → brachiocephalic artery
- **E.** Left → left subclavian; Right → right pulmonary artery

Answer: C. Left → arch of aorta; Right → proximal right subclavian artery

Why: Left 4th → aortic arch; right 4th → proximal right subclavian artery.

139. A newborn has persistence of a vascular connection between the pulmonary trunk and aorta that normally functions as a prenatal shunt. Which pharyngeal arch artery gave rise to this structure?

- **A.** Left third
- **B.** Right fourth
- **C.** Left fourth
- **D.** Distal left sixth
- **E.** Proximal right sixth

Answer: D. Distal left sixth

Why: The prenatal pulmonary-trunk-to-aorta shunt (ductus arteriosus) comes from the **distal left 6th** arch artery.

140. A developmental defect affects only the distal portion of the right sixth pharyngeal arch artery. Which adult structure would most likely be absent as a direct result?

- **A.** Right pulmonary artery
- **B.** Left pulmonary artery
- **C.** Ductus arteriosus
- **D.** No major adult structure
- **E.** Right subclavian artery

Answer: D. No major adult structure

Why: The distal right 6th arch artery **normally degenerates**, so losing it removes no adult vessel.

141. A newborn infant has difficulty breathing and swallowing. Radiographic examination reveals constrictions of both the trachea and esophagus in the same plane. Which of the following congenital malformations is the most likely cause of this condition?

A. Double aortic arch

B. Persistent right 2nd aortic arch

C. Coarctation of the aorta

D. Patent ductus arteriosus

E. Persistent left 4th aortic arch

Answer: A. Double aortic arch

Why: A persistent right dorsal aorta forms a **double aortic arch**, a vascular ring squeezing trachea and esophagus at one level.

142. How are the pharyngeal arch arteries arranged in the early embryo?

A. Six complete pairs remain simultaneously and persist unchanged into adulthood

B. Paired veins connect the sinus venosus to the dorsal aortae

C. Paired channels connect the ventral aortic sac to dorsal aortae and remodel sequentially

D. Unpaired channels connect the pulmonary veins to the aortic sac

Answer: C. Paired channels connect the ventral aortic sac to dorsal aortae and remodel sequentially

Why: Arch arteries are **paired channels** from the **aortic sac to the dorsal aortae** that appear and **remodel one after another**, never all six at once.

Intersegmental, vitelline and umbilical arteries

143. Which of the following fetal structures listed below is paired with its correct adult remnant?

A. Vitelline vein: Ligamentum teres hepatis

B. Ductus arteriosus : Ligamentum venosus

C. Sinus venosus : Left atrium

D. Umbilical artery : Medial umbilical ligament

E. 5th arch artery : Ascending aorta

Answer: D. Umbilical artery : Medial umbilical ligament

Why: Distal **umbilical arteries** become the medial umbilical ligaments; the umbilical vein (not vitelline) becomes the ligamentum teres.

Fetal versus neonatal circulation

144. As a third year medical student on your Ob/Gyn rotation, your attending asks you about fetal blood circulation. Which vessel is the most oxygenated in fetal circulation, with about 80% oxygen saturation?

A. Umbilical Vein

B. Umbilical Arteries

C. Inferior vena cava

D. Right atrium

E. Ductus arteriosus

Answer: A. Umbilical Vein

Why: The **umbilical vein**, fresh from the placenta, is the most oxygenated fetal vessel (~80%).

145. In the fetus, which of the following structures carries the lowest O2-containing blood?

A. Abdominal aorta

B. Superior Vena Cava

C. Ductus venosus

D. Umbilical vein

E. Right ventricle

Answer: B. Superior Vena Cava

Why: The **SVC** returns used blood from head and arms with no placental blood mixed in, so it has the lowest O2.

146. A medical student is asked to trace the pathway of oxygenated blood from the placenta through the fetal heart and back to the placenta. Which of the following correctly describes the pathway?

A. Placenta → Umbilical arteries → Ductus venosus → IVC → Right atrium → Right ventricle → Pulmonary arteries → Lungs

B. Placenta → Umbilical vein → Ductus venosus → IVC → Right atrium → Foramen ovale → Left atrium → Left ventricle → Aorta → Body → Umbilical arteries → Placenta

C. Placenta → Umbilical vein → Ductus arteriosus → Pulmonary artery → Right atrium → Left atrium → Aorta → Placenta

D. Placenta → Umbilical vein → IVC → Right atrium → Tricuspid valve → Right ventricle → Aorta → Umbilical arteries → Placenta

Answer: B. Placenta → Umbilical vein → Ductus venosus → IVC → Right atrium → Foramen ovale → Left atrium → Left ventricle → Aorta → Body → Umbilical arteries → Placenta

Why: Placenta → **umbilical vein** → **ductus venosus** → **IVC** → **RA** → **foramen ovale** → **LA** → **LV** → **aorta** → umbilical arteries → placenta.

147. In utero, what is the primary functional significance of the blood streaming through the ductus venosus?

A. It mixes all oxygenated placental blood uniformly with the fetal hepatic venous drainage before reaching the heart.

B. It increases pulmonary vascular resistance by directing volume overload into the right ventricle.

C. It preferentially directs well-oxygenated, high-velocity blood toward the foramen ovale to supply the brain and coronary circulation.

D. It serves as the main conduit for returning metabolic waste products directly back to the maternal circulation.

Answer: C. It preferentially directs well-oxygenated, high-velocity blood toward the foramen ovale to supply the brain and coronary circulation.

Why: Fast ductus venosus blood streams across the IVC **toward the foramen ovale**, delivering the best blood to brain and heart.

148. In fetal circulation, blood entering the right ventricle is ejected into the pulmonary trunk. Because the fetal lungs are not yet functioning as the site of gas exchange, much of this blood is redirected into which vessel?

A. Superior vena cava

B. Pulmonary veins

C. Ascending aorta

D. Descending aorta

E. Umbilical vein

Answer: D. Descending aorta

Why: Most RV output goes through the ductus arteriosus into the **descending aorta** (it joins beyond the left subclavian).

149. A 4-month-old baby is brought to the emergency room in respiratory distress. After hearing a heart murmur and observing tachycardia and bounding peripheral pulses, the physician orders an echocardiogram. Pulmonary edema is found along with a widened pulse pressure. A patent ductus arteriosus is diagnosed. This is a shunt that connects the aorta to

A. the pulmonary arteries

B. the superior vena cava

C. the pulmonary veins

D. the liver

E. the left ventricle

Answer: A. the pulmonary arteries

Why: A PDA connects the aorta to the **pulmonary arteries** (pulmonary trunk/left pulmonary artery).

150. Two diagrams show the circulation immediately before and after birth. Which combination of findings would identify the fetal circulation rather than the neonatal circulation?

A. Closed foramen ovale + ligamentum arteriosum + no placental circulation

B. Patent foramen ovale + ductus arteriosus + umbilical circulation

C. Ligamentum venosum + ductus arteriosus + closed foramen ovale

D. Four pulmonary veins + ligamentum arteriosum + umbilical vein

E. Increased left atrial pressure + patent foramen ovale + placental circulation

Answer: B. Patent foramen ovale + ductus arteriosus + umbilical circulation

Why: Before birth the foramen ovale and **ductus arteriosus are open** and the **umbilical circulation** is running.

151. A healthy infant is delivered at 39 weeks of gestation. Immediately following delivery, the infant takes its first deep breaths, cries vigorously, and the umbilical cord is clamped. Over the next several days, profound physiological shifts take place within the systemic and pulmonary vasculature to transition into adult-type circulation patterns. Which of the following correctly pairs a fetal vascular structure with its normal postnatal fibrous anatomical remnant?

A. Left umbilical vein → Ligamentum venosum

B. Ductus venosus → Ligamentum teres hepatis

C. Ductus arteriosus → Ligamentum arteriosum

D. Foramen ovale → Ostium secundum

E. Umbilical arteries → Medial umbilical folds (containing the urachus)

Answer: C. Ductus arteriosus → Ligamentum arteriosum

Why: Ductus arteriosus → ligamentum arteriosum; umbilical vein → ligamentum teres, ductus venosus → ligamentum venosum, foramen ovale → fossa ovalis.

152. A full-term newborn takes its first breath. Which of the following changes most directly promotes functional closure of the foramen ovale?

A. Increased pulmonary vascular resistance and decreased left atrial pressure

B. Decreased pulmonary blood flow and increased right atrial pressure

C. Increased pulmonary venous return and increased left atrial pressure

D. Decreased systemic vascular resistance and increased right atrial pressure

Answer: C. Increased pulmonary venous return and increased left atrial pressure

Why: First breaths raise **pulmonary venous return**, so **LA pressure** rises above RA and pushes septum primum shut.

153. Failure of which fetal vessel to close allows portal blood to bypass the liver and may cause hyperammonemia?

A. Foramen ovale

B. Ductus arteriosus

C. Ductus venosus

D. Umbilical artery

Answer: C. Ductus venosus

Why: A patent **ductus venosus** lets portal blood skip the liver, so ammonia is not cleared.

Heart tube formation and embryonic folding

154. During embryonic folding, which of the following correctly describes the positional change of the developing heart and septum transversum?

A. The heart tube moves dorsal to the foregut, and the septum transversum moves cranially to it

B. The heart tube moves ventrally, and the septum transversum comes to lie caudal to the heart

C. Both the heart tube and septum transversum remain in their original cranial position

D. The septum transversum separates the pleural and pericardial cavities from the start

Answer: B. The heart tube moves ventrally, and the septum transversum comes to lie caudal to the heart

Why: After head folding the heart lies **ventral** to the foregut with the septum transversum **caudal** to it.

155. Which embryological process brings the right and left cardiogenic regions toward the midline, allowing them to fuse and form a single primitive heart tube?

A. Cephalocaudal folding

B. Neural tube closure

C. Lateral folding

D. Septation of the primitive heart

Answer: C. Lateral folding

Why: Lateral folding brings the paired cardiogenic regions to the midline to fuse into one tube.

156. The primitive heart tube develops from the primary and secondary heart fields. Which of the following correctly describes the contribution of the primary heart field?

A. It forms most of the right ventricle and the outflow tract.

B. It forms most of the left ventricle and parts of both atria.

C. It forms the entire interventricular septum.

D. It forms only the right atrium and the pulmonary trunk.

E. It forms the valves of the heart but not the chambers.

Answer: B. It forms most of the left ventricle and parts of both atria.

Why: The **primary field** builds most of the left ventricle and parts of both atria; the secondary field adds RV and outflow tract.

157. What best describes the role of the secondary heart field during the cardiac development?

A. It forms the single endocardial tube

B. It is responsible for the organization of cardiac cells into a crescent

C. It adds progenitor cells that elongate the developing heart tube

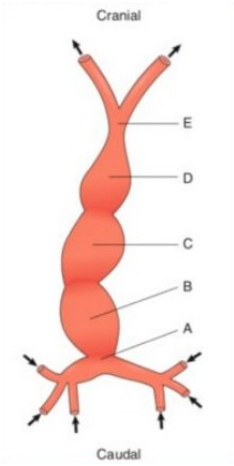
D. It establishes the initial heart tube

Answer: C. It adds progenitor cells that elongate the developing heart tube

Why: The **secondary heart field** adds progenitor cells at the poles that lengthen the heart tube.

Heart looping, segments and dextrocardia

158. Which cardiac structures derive from the heart tube dilation labeled “B”



A. Ventricles

B. Great vessels

C. Atria

D. Inferior vena cava

E. Superior vena cava

Answer: C. Atria

Why: Counting from the caudal end, A is the sinus venosus and **B is the primordial atrium**, which forms the atria.

159. You are working in the pediatric ICU and are going through the embryology of the heart with the doctor. She asks at 28 days what does the heart look like? You say it consists of:

A. The heart tube with no polarity

B. The common atrium, common ventricle, and common outflow tract

C. The common left and right atrium and common ventricle

D. The heart is fully formed

Answer: B. The common atrium, common ventricle, and common outflow tract

Why: By day 28 the looped tube has a **common atrium, common ventricle and common outflow tract**, not yet divided.

160. Match the primitive ventricle, proximal bulbus cordis, conus cordis, and truncus arteriosus to their definitive derivatives.

A. Primitive ventricle → trabeculated RV; proximal bulbus → trabeculated LV; conus → smooth ventricular outlets; truncus → proximal great arteries

B. Primitive ventricle → trabeculated LV; proximal bulbus → trabeculated RV; conus → smooth ventricular outlets; truncus → proximal great arteries

C. Primitive ventricle → trabeculated LV; proximal bulbus → atrial auricles; conus → coronary sinus; truncus → pulmonary veins

D. Primitive ventricle → smooth atria; proximal bulbus → trabeculated RV; conus → aortic arch; truncus → ventricular septum

Answer: B. Primitive ventricle → trabeculated LV; proximal bulbus → trabeculated RV; conus → smooth ventricular outlets; truncus → proximal great arteries

Why: Primitive ventricle → trabeculated LV; **proximal bulbus → trabeculated RV**; conus → smooth outlets (infundibulum, aortic vestibule); truncus → proximal aorta and pulmonary trunk.

161. Which of the following embryonic structures gives rise to the smooth part of the right ventricle (conus arteriosus)?

A. Primitive Ventricle

B. Primitive Atrium

C. Bulbus Cordis

D. Sinus Venosus

E. Truncus Arteriosus

Answer: C. Bulbus Cordis

Why: The **bulbus cordis** becomes the smooth conus arteriosus of the RV (and aortic vestibule of the LV).

162. During heart development, what happens during heart tube looping?

A. The heart tube divides into two separate hearts.

B. The heart tube becomes part of the developing lungs.

C. The heart tube forms the diaphragm.

D. The heart tube stops growing and begins to degenerate.

E. The heart tube bends and twists, positioning the future heart chambers correctly.

Answer: E. The heart tube bends and twists, positioning the future heart chambers correctly.

Why: During looping the tube **bends and twists** to bring the future chambers into their final positions.

163. During normal rightward looping, where do the bulbus cordis, primitive ventricle, and atrial region move?

- **A.** Bulbus moves ventrally caudally and leftward; ventricle shifts right; atria move dorsally and cranially
- **B.** Bulbus moves dorsally and cranially; ventricle shifts left; atria move ventrally and caudally
- **C.** Bulbus and ventricle remain fixed; only the atria move rightward
- **D.** Bulbus moves ventrally caudally and rightward; ventricle shifts left; atria move dorsally and cranially

Answer: D. Bulbus moves ventrally caudally and rightward; ventricle shifts left; atria move dorsally and cranially

Why: In rightward looping the **bulbus cordis moves ventrally, caudally and to the right**, the **ventricle shifts left**, and the **atria move dorsally and cranially**.

164. A couple brings their 5-day-old baby for its first checkup after birth. The doctor listens to the baby's heart and notices that the heartbeat is much louder on the right side then the left side. What condition does the baby most likely have?

- **A.** nothing, this is completely normal
- **B.** an atrial septal defect
- **C.** a ventricular septal defect
- **D.** dextrocardia
- **E.** Tetralogy of Fallot

Answer: D. dextrocardia

Why: Heart sounds loudest on the right mean the heart points right: **dextrocardia** from abnormal looping.

AV canal and atrial septation

165. During partitioning of the heart the development of which of the following structures leads to the formation of the left and right atrioventricular canals?

- **A.** Outflow tract
- **B.** Endocardial cushions
- **C.** Interventricular septum
- **D.** Pulmonary truncus
- **E.** Right sinual horn

Answer: B. Endocardial cushions

Why: Dorsal and ventral **endocardial cushions** fuse and split the AV canal into right and left canals.

166. Which of the following best details the correct sequence of events leading to the partitioning of the atrium?

- **A.** Fusion of endocardial cushions ? Septum primum? Foramen secundum ? Foramen primum
- **B.** Septum primum? Foramen primum ? Septum secundum ? Fusion of endocardial cushions
- **C.** Septum primum ? Foramen primum ? Foramen secundum ? Septum secundum
- **D.** Foramen primum ? Septum secundum ? Foramen secundum ? Septum primum
- **E.** Septum primum ? Septum secundum ? Foramen primum ? Foramen secundum

Answer: C. Septum primum ? Foramen primum ? Foramen secundum ? Septum secundum

Why: Septum primum → foramen primum → foramen secundum (perforations) → **septum secundum**.

167. A male baby is born with Down syndrome (trisomy 21) and associated cardiac defects. The given apical four-chamber view echocardiogram shows a complete atrioventricular canal defect that includes a primum atrial septal defect and a posterior inlet ventricular septal defect. What embryonic structures normally fuse to close the foramen primum?

- **A.** Septum secundum and the septum primum
- **B.** Septum secundum and the fused endocardial cushions
- **C.** Septum primum and the fused endocardial cushions
- **D.** Septum primum and the septum spurium
- **E.** Left inferior and right superior truncus swellings

Answer: C. Septum primum and the fused endocardial cushions

Why: Foramen primum closes when the **septum primum** fuses with the **fused endocardial cushions**.

168. You are in open lab and going through the structures of the heart. Your fellow classmate points the probe at the fossa ovalis and asks you “how does this thing even close in a baby”? You answer:

- **A.** Foramen ovale and foramen secundum fuse upon first breath
- **B.** Septum primum and foramen primum fuse upon first breath
- **C.** Septum secundum and foramen secundum fuse upon first breath
- **D.** Septum primum and foramen ovale fuse upon first breath
- **E.** Septum primum and septum secundum fuse upon first breath

Answer: E. Septum primum and septum secundum fuse upon first breath

Why: The fossa ovalis forms when **septum primum** (the valve) fuses with **septum secundum** after birth.

Atrial septal defects

169. During a routine wellness visit, the pediatrician notices a slight murmur on auscultation of an 8-month-old male. A follow-up echocardiogram demonstrates the presence of a probe patent foramen ovale. The pediatrician tells the mother that they will continue monitoring the child but that there is no need to worry. Which two structures failed to properly fuse in this patient after birth?

A. Foramen secundum and septum primum

B. Foramen ovale and foramen primum

C. Foramen ovale valve and septum secundum

D. Foramen ovale and endocardial cushion

E. Foramen primum and foramen secundum

Answer: C. Foramen ovale valve and septum secundum

Why: In probe PFO the **valve of the foramen ovale** (septum primum) never seals to **septum secundum**.

170. A 25-year-old rancher is admitted to the hospital after having a mild stroke. History of the rancher includes birth control and smoking. Blood oxygen is slightly low. An echocardiogram indicates no sign of a ventricular septal defect or a patent ductus arteriosus but does indicate lower than normal pressure in the left atrium. The condition that the rancher has is

A. patent foramen ovale

B. transposition of the great arteries

C. Tetralogy of Fallot

D. Coarctation of the aorta

Answer: A. patent foramen ovale

Why: A young stroke with low LA pressure fits a venous clot crossing a **patent foramen ovale** (paradoxical embolism).

171. An 18-year-old patient presents with a history of exercise intolerance, repeated bouts of pneumonia, and pulmonary hypertension. Testing reveals that the patient is getting varying amounts of blood pumped to her lungs. An echocardiogram indicates a normal interventricular septum and a normal fossa ovalis. The patient is diagnosed with

A. Transposition of the great arteries

B. An interventricular septal defect

C. An interatrial septal defect

D. Lazyness

E. Dextrocardia

Answer: C. An interatrial septal defect

Why: Normal IV septum but a shunt to the lungs with a normal fossa = an **interatrial** defect elsewhere in the septum.

172. A 4-year-old girl is brought to the pediatrician for a routine checkup. On cardiac auscultation, the physician notes a fixed, widely split-second heart sound (S2) and a mild midsystolic ejection murmur at the upper left sternal border. An echocardiogram reveals an abnormal communication between the right and left atria resulting from excessive resorption of the septum primum. Failure of which embryological structure to properly develop or overlap best characterizes the most common class of this congenital defect?

A. Septum transversum

B. Septum secundum

C. Muscular interventricular septum

D. Spiral aorticopulmonary septum

E. Left sixth aortic arch

Answer: B. Septum secundum

Why: Secundum ASD: too much septum primum resorbed and/or a **septum secundum** too short to overlap the foramen secundum.

173. A 2-day-old neonate is severely cyanotic and has difficulty breathing. A cardiovascular exam reveals a large opening in the atrial septum. The improper overlapping of which of the following leads to this congenital anomaly?

A. Foramen secundum and septum primum

B. Foramen ovale and foramen primum

C. Foramen ovale and septum secundum

D. Foramen ovale and foramen secundum

E. Foramen primum and foramen secundum

Answer: D. Foramen ovale and foramen secundum

Why: A large secundum ASD forms when the **foramen ovale and foramen secundum** line up instead of each being covered by the other septum.

174. Failure of proper development of the sinus venosus is a cause associated with which of the following defects?

A. Ventriculoseptal defects

B. Patent ductus arteriosus

C. Persistent ductus venosus

D. Agenesis of the inferior vena cava

E. Atrioseptal defects

Answer: E. Atrioseptal defects

Why: Faulty absorption of the **sinus venosus** causes the sinus venosus type of **atrial septal defect**.

Sinus venosus fate and definitive atria

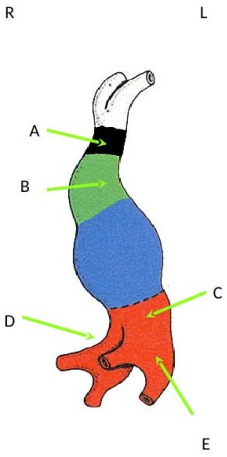
175. The coronary sinus is derived from which of the following?

A. Ductus venosus
B. Bulbus cordis
C. Primitive ventricle
D. Right sinus venosus
E. Left sinus venosus

Answer: E. Left sinus venosus

Why: The **left horn of the sinus venosus** becomes the coronary sinus.

176. An MRI of a month old female is recorded after having several respiratory issues. The scan reveals no issues with her bronchial tree but it does reveal an enlarged coronary sinus valve opening into the right atrium. Which of the following structures of the embryologic heart tube gives rise to the coronary sinus?



A. A
B. B
C. C
D. D
E. E

Answer: E. E

Why: Arrow E marks the **left sinus horn**, which becomes the coronary sinus.

177. During embryologic development, the right horn of the sinus venosus becomes incorporated into the right atrium. Which of the following adult structures derives from this region?

A. Coronary sinus
B. Oblique vein of the left atrium
C. Smooth part of the right atrial wall (sinus venarum)
D. Rough, trabeculated part of the right atrium
E. Crista terminalis

Answer: C. Smooth part of the right atrial wall (sinus venarum)

Why: The right sinus horn is absorbed as the smooth **sinus venarum** of the right atrium.

178. A researcher observes that the primitive pulmonary vein does not become incorporated into the wall of the developing left atrium. Which adult atrial region would be most directly affected?

A. Trabeculated left atrial appendage
B. Smooth-walled portion of the left atrium
C. Smooth-walled portion of the right atrium
D. Coronary sinus
E. Interatrial septum

Answer: B. Smooth-walled portion of the left atrium

Why: The absorbed primitive pulmonary vein forms the **smooth-walled left atrium**; the auricle is primitive atrium.

179. Why does each atrium contain both a smooth region and a trabeculated region?

A. Primitive atrium forms smooth regions; right sinus horn forms trabeculated RA; pulmonary vein incorporation forms trabeculated LA
B. Primitive atrium forms trabeculated regions; right sinus horn forms smooth RA; pulmonary vein incorporation forms smooth LA
C. Primitive atrium forms trabeculated regions; left sinus horn forms smooth LA; right umbilical vein forms smooth RA
D. Primitive ventricle forms trabeculated atria; cardinal veins form both smooth atrial walls

Answer: B. Primitive atrium forms trabeculated regions; right sinus horn forms smooth RA; pulmonary vein incorporation forms smooth LA

Why: Trabeculated parts come from the **primitive atrium**; the smooth RA comes from the **right sinus horn** and the smooth LA from **incorporated pulmonary veins**.

Ventricular septation and VSDs

180. During normal cardiac development, the interventricular septum forms through the fusion of several embryological structures. A defect in this process may lead to a ventricular septal defect. Which of the following correctly describes the formation of the interventricular septum?

A. It develops entirely from the muscular ridge growing upward from the primitive ventricle.

B. It forms from the downward growth of the endocardial cushions alone.

C. It develops from both a muscular portion arising from the floor of the primitive ventricle and a membranous portion formed by the fusion of the endocardial cushions, aorticopulmonary septum, and muscular septum.

D. It develops from the fusion of the right and left atrioventricular canals.

E. It forms solely from the aorticopulmonary septum dividing the truncus arteriosus

Answer: C. It develops from both a muscular portion arising from the floor of the primitive ventricle and a membranous portion formed by the fusion of the endocardial cushions, aorticopulmonary septum, and muscular septum.

Why: Muscular septum from the ventricle floor plus a **membranous** part from cushions, AP septum and muscular septum.

181. A 2-month-old infant is evaluated for poor feeding, sweating during feeds, and failure to thrive. On physical examination, a harsh, blowing holosystolic murmur is loudest at the left lower sternal border. Echocardiography demonstrates a defect in the membranous portion of the interventricular septum. A failure of which of the following embryonic structures to properly proliferate and fuse is the primary cause of this specific type of defect?

A. Primitive atrium and sinus venosus

B. Right and left lateral pharyngeal arches

C. Endocardial cushions

D. Splanchnic mesoderm of the right venous valve

E. Neural crest cells forming the truncus arteriosus wall exclusively

Answer: C. Endocardial cushions

Why: The membranous IV septum comes mainly from **endocardial cushion** tissue (with AP septum), so its defect is a cushion failure.

182. A faulty fusion of the right and left bulbar ridges and AV cushion will result in which of the following?

A. Persistent truncus arteriosus (PTA)

B. Ebstein's anomaly

C. Transposition of the great arteries

D. Common ventricle

E. Membranous VSD

Answer: E. Membranous VSD

Why: The membranous septum forms where bulbar ridges and AV cushions meet; faulty fusion leaves a **membranous VSD**.

183. A ventricular septal defect (VSD) most commonly occurs in which portion of the interventricular septum?

A. Muscular portion near the apex

B. Membranous portion near the atrioventricular valves

C. Interatrial septum near the foramen ovale

D. Septum between the pulmonary trunk and aorta

Answer: B. Membranous portion near the atrioventricular valves

Why: Most VSDs are in the **membranous** septum near the AV valves.

184. An infant is found to have an opening in the superior portion of the interventricular septum, near the septal leaflet of the tricuspid valve. Which portion of the septum is most likely involved?

A. Muscular portion near the apex

B. Membranous portion

C. Interatrial septum

D. Crista terminalis

Answer: B. Membranous portion

Why: An upper-septum hole by the septal tricuspid leaflet is **membranous** (perimembranous) VSD.

185. A 10-year-old child is diagnosed with mild congestive heart failure accompanied by mild pulmonary arterial hypertension. An echocardiogram indicates a small shunt between the two ventricles. The condition that causes this is

A. patent foramen ovale

B. atrial septal defect

C. ventricular septal defect

D. failure of heart looping

E. Tetralogy of Fallot

Answer: C. ventricular septal defect

Why: A shunt **between the ventricles** is a ventricular septal defect.

Outflow tract and transposition

186. During embryonic development, neural crest cells migrate into the cardiac outflow tract and contribute to the formation of the conotruncal ridges. What is the primary function of these ridges?

A. Separate the atria into right and left chambers

B. Divide the truncus arteriosus into the ascending aorta and pulmonary trunk

C. Form the muscular portion of the interventricular septum

D. Separate the pulmonary veins from the systemic veins

Answer: B. Divide the truncus arteriosus into the ascending aorta and pulmonary trunk

Why: Conotruncal ridges fuse into the AP septum that divides the truncus into **aorta and pulmonary trunk**.

187. A student examining sequential sections through the developing outflow tract notices that the relative positions of the aortic and pulmonary channels change between superior and inferior sections. Which developing structure accounts for this arrangement?

A. Aorticopulmonary septum

B. Interventricular septum

C. Septum primum

D. Endocardial cushions

E. Sinus venosus

Answer: A. Aorticopulmonary septum

Why: The **spiral** aorticopulmonary septum makes aorta and pulmonary trunk twist around each other.

188. A newborn is diagnosed with persistent truncus arteriosus due to abnormal development of the cardiac outflow tract. Dysfunction of which embryological cell population is most directly associated with this defect?

A. Endocardial cushion cells

B. Somatic mesoderm cells

C. Neural crest cells

D. Septum transversum cells

Answer: C. Neural crest cells

Why: Persistent truncus results when **neural crest** cells fail to build the AP septum.

189. An aorticopulmonary septal defect would cause blood to flow or mix in which way?

A. blood would mix between atria

B. blood would mix between ventricles

C. blood would flow to lungs instead of the body through the aorta and the body instead of the lungs through the pulmonary trunk

D. blood would flow to the body instead of the lungs through the aorta

E. blood would mix between the aorta and the pulmonary trunk

Answer: E. blood would mix between the aorta and the pulmonary trunk

Why: With no AP septum the truncus stays one vessel, so blood **mixes between aorta and pulmonary trunk**.

190. At birth, an infant is diagnosed with the very severe condition of transposition of the great arteries. Prostaglandin (Alprostadil – PGE1) is administered in order to keep the ductus arteriosus patent. As soon as the patient is stable, surgery is performed to correct the problem. Transposition of the great arteries means that

A. the aorta arises from the left ventricle

B. the aorta arises from the right ventricle

C. the aorta arises from the left atrium

D. the aorta arises from the right atrium

Answer: B. the aorta arises from the right ventricle

Why: In TGA the **aorta arises from the right ventricle** (and the pulmonary trunk from the left).

191. A neonate is rushed to the intensive care neonate unit shortly after delivery with cyanosis and mild respiratory distress. A CT scan showed the clinical presentation of transposition of the great arteries. Which of the following defects leads to this condition?

A. Abnormal resorption of the septum primum

B. Failure of fusion of the endocardial cushions

C. Faulty partitioning of the truncus arteriosus

D. Involution of the ductus arteriosus

E. Persistence of the 5th arch arteries

Answer: C. Faulty partitioning of the truncus arteriosus

Why: TGA comes from **faulty partitioning of the truncus**: the AP septum fails to spiral.

192. A newborn infant develops profound, generalized cyanosis within hours of birth that does not improve with supplemental oxygen administration. Echocardiography reveals that the aorta arises directly from the right ventricle and the pulmonary trunk arises from the left ventricle. A patent ductus arteriosus is noted, which is currently preserving life by allowing mixing of the parallel circulatory loops. Failure of which embryological process is responsible for this life-threatening defect?

A. Degeneration of the right vitelline and umbilical venous systems

B. Clockwise dextrolooping of the primary endocardial heart tube

C. Involution of the left fourth pharyngeal arch artery

D. Proper migration and differentiation of cardiac neural crest cells within the outflow tract

E. Ingrowth of the muscular interventricular septum from the apex

Answer: D. Proper migration and differentiation of cardiac neural crest cells within the outflow tract

Why: TGA follows failed **neural crest** migration into the outflow tract, so the AP septum does not spiral.

Other malformations and clinical integration

193. A physician orders an oxygen saturation exam for a neonate who was monitored for cardiovascular development issues throughout pregnancy. All levels look normal, except for the oxygen level in the aorta which is much lower. What is this an indication of?

A. Atrial septal defect
B. Patent ductus arteriosus
C. Ventricular septal defect
D. Patent ductus venosus
E. Patent umbilical arteries

Answer: B. Patent ductus arteriosus

Why: Only the aorta desaturated (LA, LV normal) means venous blood enters beyond the LV: right-to-left flow through a **PDA**.

194. A 3-week-old female infant born prematurely at 30 weeks of gestation is evaluated in the neonatal intensive care unit for worsening respiratory distress, hyperdynamic precordium, and bounding peripheral pulses. On physical examination, a harsh, continuous "machinery-like" murmur is heard loudest at the left upper sternal border. An echocardiogram confirms that a major fetal circulatory shunt, which normally closes postnatally in response to increased oxygen tension and falling prostaglandin levels, has failed to constrict. Which of the following congenital vascular anomalies best describes this condition?

A. Coarctation of the aorta
B. Transposition of the great vessels
C. Persistent ductus arteriosus
D. Total anomalous pulmonary venous return
E. Persistent truncus arteriosus

Answer: C. Persistent ductus arteriosus

Why: Premature baby, continuous machinery murmur, bounding pulses = a **ductus arteriosus** that failed to constrict.

195. During a newborn examination, a pediatrician is unable to detect a pulse in the groin or legs of an infant and notes the lower extremities are cold to the touch. The pulse and blood pressure of the upper limbs are significantly elevated from normal. A pediatric cardiologist performs an echocardiogram and other imaging. The given sagittal MRI verifies the presence of what specific anomaly, as indicated by the white arrowhead?

A. Coarctation of the aorta
B. Tetralogy of Fallot
C. Ventricular septal defect (VSD)
D. Atrial septal defect (ASD)
E. Transposition of the great vessels

Answer: A. Coarctation of the aorta

Why: Absent leg pulses, cold legs and high arm pressure = **coarctation of the aorta**.

196. A 4-year-old boy presents with hypertension in the upper extremities and a diminished femoral pulse and pressure. Radiologic imaging reveals a postductal coarctation of the aorta. Which of the following is the most characteristic feature of this condition?

A. Patent ductus arteriosus
B. Closed ductus arteriosus
C. Patent ductus venosus
D. Patent foramen ovale
E. Stenotic aortic valve

Answer: B. Closed ductus arteriosus

Why: In postductal (adult) coarctation the **ductus is closed** and collaterals feed the lower body.

197. A child has pulmonary outflow obstruction, a VSD, an overriding aorta, and right ventricular hypertrophy. What developmental change links these findings?

A. Posterior displacement of septum primum; RV hypertrophy is a primary atrial septation defect
B. Complete absence of conotruncal partitioning; a common valve causes all four findings
C. Failure of left fourth arch remodeling; isolated aortic narrowing directly causes the VSD
D. Anterosuperior malalignment of the outlet septum; RV hypertrophy follows pressure loading

Answer: D. Anterosuperior malalignment of the outlet septum; RV hypertrophy follows pressure loading

Why: Tetralogy comes from **anterosuperior malalignment of the outlet septum**; **RV hypertrophy is secondary** to pressure load.

198. At birth a doctor notices that a baby is severely cyanotic. A chest X-ray indicates a boot-shaped heart, and the diagnosis of Tetralogy of Fallot is made. Why is the heart boot-shaped?

A. Hypertrophy of the left atrium
B. Hypertrophy of the right atrium
C. Hypertrophy of the left ventricle
D. Hypertrophy of the right ventricle

Answer: D. Hypertrophy of the right ventricle

Why: The upturned apex of the boot comes from **right ventricular hypertrophy**.

199. During the initial checkup of a neonate, the physician notices an odd murmur when auscultating the aortic and pulmonary valves. Further investigation reveals that both the ascending aorta and pulmonary trunk are communicating with both ventricles leading to a mixture of oxygenated and deoxygenated blood to be pumped into the pulmonary and systemic circulation. A diagnosis of persistent truncus arteriosus is made. What other congenital anomaly will be observed in the heart of this neonate?

A. Pulmonary stenosis
B. Coarctation of the aorta
C. Patent ductus venosus
D. Atrial septal defect
E. Ventricular septal defect

Answer: E. Ventricular septal defect

Why: Persistent truncus arteriosus is always accompanied by a **VSD**.

200. Which option correctly distinguishes transposition, persistent truncus, and Tetralogy of Fallot?

- **A.** Transposition: common outlet; truncus: wrong ventricular connections; Tetralogy: isolated atrial defect
- **B.** Transposition: normal connections with PDA; truncus: two separate valves and a window; Tetralogy: aortic isthmus narrowing
- **C.** Transposition: wrong ventricular connections; truncus: common outlet; Tetralogy: malalignment with RV outflow obstruction
- **D.** Transposition: isolated VSD; truncus: isolated pulmonary stenosis; Tetralogy: failure of atrial flap fusion

Answer: C. Transposition: wrong ventricular connections; truncus: common outlet; Tetralogy: malalignment with RV outflow obstruction

Why: Transposition = wrong ventricular connections; **truncus** = one common outlet; **tetralogy** = malalignment with **RV outflow obstruction**.